42

Atomic Physics

CHAPTER OUTLINE

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* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ42.1 **(i)** Answer (e). **(ii)** Answer (c). The M means that the electron falls into the M shell, for which $n = 3$. The β means the electron comes from two shells above M: the O shell, for which $n = 5$. M_α would refer to $4 \rightarrow 3$ and M_β refers to $5 \rightarrow 3$.

OQ42.2 Answer (c). All states associated with $\ell = 2$ are referred to as d states. Thus, all 10 possible quantum states having $n = 3$, $\ell = 2$ are called 3d states.

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OQ42.3 Answer (d). Wavelengths of the hydrogen spectrum are given by $1/\lambda = R_{\mathrm{H}}\left(1/n_f^2 - 1/n_i^2\right)$, where R_{H} is the Rydberg constant. For the transition $n_i = 5$ to $n_f = 3$, we have

$$\frac{1}{\lambda} = \left(1.097 \times 10^7 \text{ m}^{-1}\right)\left(\frac{1}{3^2} - \frac{1}{5^2}\right) = 7.80 \times 10^5 \text{ m}^{-1}$$

$$\lambda = 1.28 \times 10^{-6} \text{ m}$$

OQ42.4 Answer (e). With a principal quantum number of $n = 3$, there are 3 possible values of the orbital quantum number, $\ell = 0,\ 1,\ 2.$ There are a total of $2(2\ell + 1)$ possible quantum states for each value of ℓ, $2\ell + 1$ possible values of the orbital magnetic quantum number m_ℓ, and 2 possible spin orientations $(m_s = \pm\frac{1}{2})$ for each value of m_ℓ.

Thus, the number of states are

$3s$ states $(n = 3, \ell = 0)$: 2[2(0) + 1] = 2

$3p$ states $(n = 3, \ell = 1)$: 2[2(1) + 1] = 6

$3d$ states $(n = 3, \ell = 2)$: 2[2(2) + 1] = 10

The grand total of $n = 3$ states is 2 + 6 + 10 = 18.

OQ42.5 Answer (c). It is an experimental fact the charge on the electron is quantized. The Bohr model does not introduce this as a new assumption.

OQ42.6 **(i)** Answer (b). **(ii)** Answer (e). From the discussion of Equations 42.8 and 42.9, $K = \frac{k_e e^2}{2r}$ and $U_e = -\frac{k_e e^2}{r}$. If

$$-E = K + U_e = +\frac{k_e e^2}{2r} - \frac{k_e e^2}{r} = -\frac{k_e e^2}{2r}, \text{ then } E = \frac{k_e e^2}{2r}.$$

Therefore, $K = E$ and $U_e = -2E$.

OQ42.7 Answer (e). The structure of the periodic table is the result of the Pauli exclusion principle, which states that no two electrons in an atom can ever have the same set of values for the set of quantum numbers $n,\ \ell,\ m_\ell,$ and m_s.

OQ42.8 (a) Yes, provided that the energy of the photon is *precisely* enough to put the electron into one of the allowed energy states. Strangely—more precisely non-classically—enough, if the energy of the photon is not sufficient to put the electron into a particular excited energy level, the photon will not interact with the atom at all!

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(b) Yes, a photon of any energy greater than 13.6 eV will ionize the atom. Any energy above 13.6 eV will go into kinetic energy of the newly liberated electron.

OQ42.9 Answers (b) and (e). Choice (b) is not possible because the Pauli exclusion principle limits the number of electrons in any p subshell to a maximum of 6. Choice (e) is impossible because the selection rules of quantum mechanics limit the maximum value of ℓ to $n - 1$. Thus, a $2d$ state ($n = 2,\ \ell = 2$) cannot exist.

OQ42.10 Answer (e). Since the electron is in some bound quantum state of the atom, the atom is not ionized and choice (a) is false. The fact that the electron is in a d state means that its orbital quantum number is $\ell = 2,$ so choice (b) is false. Also, since the maximum value of ℓ is $n - 1$, choice (c) is false. Finally, the ground state of hydrogen is a $1s$ state, so choice (d) is false. Choice (e) is true because the magnitude of the orbital angular momentum is $L = \sqrt{\ell(\ell+1)}\hbar = \sqrt{2(2+1)}\hbar = \sqrt{6}\hbar.$

OQ42.11 **(i)** In order of energy change, the ranking is a > d > c > b.

(ii) In order of decreasing photon wavelength, the ranking is c = d > b > a.

We calculate the energy of the photon according to $\Delta E = (13.6\text{ eV})\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)$, where $\Delta E > 0$ means the photon is absorbed and $\Delta E < 0$ means the photon is emitted. We calculate the wavelength according to $\lambda = \frac{hc}{\Delta E} = \frac{1\ 240\text{ eV}\cdot\text{nm}}{\Delta E}$.

(a) $n_i = 2$ and $n_f = 5$, $\Delta E = 2.86$ eV (absorption) $\lambda = 434$ nm

(b) $n_i = 5$ and $n_f = 3$, $\Delta E = -0.967$ eV (emission) $\lambda = 1\ 280$ nm

(c) $n_i = 7$ and $n_f = 4$, $\Delta E = -0.572$ eV (emission) $\lambda = 2\ 170$ nm

(d) $n_i = 4$ and $n_f = 7$, $\Delta E = 0.572$ eV (absorption) $\lambda = 2\ 170$ nm

OQ42.12 Answer (c). The photon carries energy, thus an electron must lose energy.

OQ42.13 (a) Yes. As $n \to \infty,\ E_n = -13.6\text{ eV}/n^2 \to 0,$ and the electron remains in a bound state.

(b) No. To produce a spectral line, the electron must make a transition from a higher energy bound state to a lower energy bound state. The greatest frequency is that of the Lyman series limit, caused by the transition from $n = \infty$ to $n = 1$.

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(c) Yes. Photons with large wavelengths, corresponding to low photon energies, can be produced by transitions between adjacent states with n large.

OQ42.14 **(i)** Answer (d). The spin quantum number $m_s = \pm 1/2$.

(ii) Answers (c) and (d). The orbital magnetic quantum number m_ℓ has the range $-\ell, -\ell+1, \ldots, -1, 0, 1, \ldots, \ell-1, \ell$, and spin quantum number $m_s = \pm 1/2$.

(iii) Answers (b) and (c). The orbital quantum number has values $\ell = 0, 1, 2, \ldots, n-1$, and, as stated above, m_ℓ can be zero.

OQ42.15 Answer (a). The bombarding electron can give up all or part of its kinetic energy to the atom. The energy required to raise the atom from its ground state to its first excited state is

$$\Delta E = E_2 - E_1 = -\frac{13.6 \text{ eV}}{2^2} - \left(-\frac{13.6 \text{ eV}}{1^2}\right) = 10.2 \text{ eV}$$

The bombarding electron can give up this energy to the atom and carry off the remaining 0.3 eV.

ANSWERS TO CONCEPTUAL QUESTIONS

CQ42.1 Stimulated emission coerces atoms to emit photons along a specific axis and in phase rather than in the random directions and phases of spontaneously emitted photons. The photons that are emitted through stimulation can be made to accumulate over time. The fraction allowed to escape constitutes the intense, collimated, and coherent laser beam. If this process relied solely on spontaneous emission, the emitted photons would not exit the laser tube or crystal in the same direction. Neither would they be coherent with one another.

CQ42.2 In a neutral helium atom, one electron can be modeled as moving in an electric field created by the nucleus and the other electron. According to Gauss's law, if the electron is above the ground state it moves in the electric field of a net charge of $+2e - 1e = +1e$. We say the nuclear charge is *screened* by the inner electron. The electron in a He^+ ion moves in the field of the unscreened nuclear charge of 2 protons. Then the potential energy function for the electron is about double that of one electron in the neutral atom.

CQ42.3 Fundamentally, three quantum numbers describe an orbital wave function because we live in three-dimensional space. They arise mathematically from boundary conditions on the wave function,

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expressed as a product of a function of r, a function of θ, and a function of ϕ.

CQ42.4 Bohr's theory pictures the electron as moving in a flat circle like a classical particle described by $\sum F = ma$. Schrödinger's theory pictures the electron as a cloud of probability amplitude in the three-dimensional space around the hydrogen nucleus, with its motion described by a wave equation. In the Bohr model, angular momentum can take the values $L = n\hbar$, $n = 1, 2, 3, \ldots$, so the ground-state angular momentum is $1\hbar$; in the Schrödinger model, angular momentum can take the values $L = \sqrt{\ell(\ell+1)}\hbar$, $\ell = 0, 1, \ldots, n-1$, so the ground-state angular momentum ($n = 1 \rightarrow \ell = 0$) is zero. Both models predict that the electron's energy is limited to discrete energy levels, given by $-13.6\ \text{eV}/n^2$, with $n = 1, 2, 3, \ldots$.

CQ42.5 Practically speaking, no. Ions have a net charge and the magnetic force $q\left(\vec{\mathbf{v}} \times \vec{\mathbf{B}}\right)$ would deflect the beam, making it difficult to separate the atoms with different orientations of magnetic moments.

CQ42.6 The deflecting force on an atom with a magnetic moment is proportional to the *gradient* of the magnetic field. Thus, atoms with oppositely directed magnetic moments would be deflected in *opposite* directions in an inhomogeneous magnetic field.

CQ42.7 If the exclusion principle were not valid, the elements and their chemical behavior would be grossly different because every electron would end up in the lowest energy level of the atom. All matter would be nearly alike in its chemistry and composition, since the shell structures of all elements would be identical. Most materials would have a much higher density. The spectra of atoms and molecules would be very simple, and there would be very little color in the world.

CQ42.8 Bohr modeled the electron as moving in a perfect circle, with zero uncertainty in its radial coordinate. Then its radial velocity is always zero with zero uncertainty. Bohr's theory violates the uncertainty principle by making the uncertainty product $\Delta r \Delta p_r$ be zero, less than the minimum allowable $\hbar/2$.

CQ42.9 The three elements have similar electronic configurations. Each has filled inner shells plus one electron in an outer *s* orbital. Their single outer electrons largely determine their chemical interactions with other atoms.

CQ42.10 Each of the electrons must have at least one quantum number different from the quantum numbers of each of the other electrons. They can differ (in m_s) by being spin-up or spin-down. They can also differ (in ℓ) in angular momentum. Those electrons with $\ell = 1$ can

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differ (in m_ℓ) in orientation of angular momentum. For $n = 2$, $\ell = 0$ or 1. If $\ell = 0$, $m_\ell = 0$, and $m_s = \pm 1/2$, for a total of two different states. For $\ell = 1$, $m_\ell = -1,\ 0, +1$, and $m_s = \pm 1/2$, for a total of six different states.

CQ42.11 If an electron moved like a hockey puck, it could have any arbitrary frequency of revolution around an atomic nucleus. If it behaved like a charge in a radio antenna, it would radiate light with frequency equal to its own frequency of oscillation. Thus, the electron in hydrogen atoms would emit a continuous spectrum, electromagnetic waves of all frequencies smeared together.

CQ42.12 No. Laser light is collimated. The energy generally travels in the same direction. The intensity of a laser beam stays remarkably constant, independent of the distance it has traveled.

SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 42.1 Atomic Spectra of Gases

P42.1 (a) The wavelengths in the Lyman series of hydrogen are given by

$$\frac{1}{\lambda} = R_{\mathrm{H}}\left(1 - \frac{1}{n^2}\right)$$

where $n = 2, 3, 4, \ldots$, and the Rydberg constant is $R_{\mathrm{H}} = 1.097\ 373\ 2 \times 10^7\ \mathrm{m}^{-1}$. This can also be written as

$$\lambda = \left(\frac{1}{R_{\mathrm{H}}}\right)\left(\frac{n^2}{n^2 - 1}\right)$$

therefore, the first three wavelengths in this series are

$$\lambda_1 = \frac{1}{1.097\ 373\ 2 \times 10^7\ \mathrm{m}^{-1}}\left(\frac{2^2}{2^2 - 1}\right) = 1.215 \times 10^{-7}\ \mathrm{m}$$

$$= \boxed{121.5\ \mathrm{nm}}$$

$$\lambda_2 = \frac{1}{1.097\ 373\ 2 \times 10^7\ \mathrm{m}^{-1}}\left(\frac{3^2}{3^2 - 1}\right) = 1.025 \times 10^{-7}\ \mathrm{m}$$

$$= \boxed{102.5\ \mathrm{nm}}$$

$$\lambda_3 = \frac{1}{1.097\ 373\ 2 \times 10^7\ \mathrm{m}^{-1}}\left(\frac{4^2}{4^2 - 1}\right) = 9.720 \times 10^{-8}\ \mathrm{m}$$

$$= \boxed{97.20\ \mathrm{nm}}$$

(b) These wavelengths are all in the ultraviolet of the spectrum.

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P42.2 (a) The wavelengths in the Paschen series of hydrogen are given by

$$\frac{1}{\lambda} = R_H\left(\frac{1}{3^2} - \frac{1}{n^2}\right)$$

where $n = 4, 5, 6, \ldots$, and the Rydberg constant is $R_H = 1.097\ 373\ 2 \times 10^7\ \text{m}^{-1}$. This can also be written as

$$\lambda = \left(\frac{1}{R_H}\right)\left(\frac{9n^2}{n^2 - 9}\right)$$

therefore, the first three wavelengths in this series are

$$\lambda_1 = \frac{1}{1.097\ 373\ 2 \times 10^7\ \text{m}^{-1}}\left[\frac{9(4)^2}{4^2 - 9}\right] = 1.875 \times 10^{-6}\ \text{m}$$

$$= \boxed{1\ 875\ \text{nm}}$$

$$\lambda_2 = \frac{1}{1.097\ 373\ 2 \times 10^7\ \text{m}^{-1}}\left[\frac{9(5)^2}{5^2 - 9}\right] = 1.281 \times 10^{-6}\ \text{m}$$

$$= \boxed{1\ 281\ \text{nm}}$$

$$\lambda_1 = \frac{1}{1.097\ 373\ 2 \times 10^7\ \text{m}^{-1}}\left[\frac{9(6)^2}{6^2 - 9}\right] = 1.094 \times 10^{-6}\ \text{m}$$

$$= \boxed{1\ 094\ \text{nm}}$$

(b) These wavelengths are all in the $\boxed{\text{infrared}}$ region of the spectrum.

P42.3 (a) The fifth excited state must lie above the second excited state by the photon energy

$$E_{52} = hf = \frac{hc}{\lambda} = \frac{\left(6.626 \times 10^{-34}\ \text{J} \cdot \text{s}\right)\left(3.00 \times 10^8\ \text{m/s}\right)}{520 \times 10^{-9}\ \text{m}}$$

$$= 3.82 \times 10^{-19}\ \text{J}$$

The sixth excited state exceeds the second in energy by

$$E_{62} = \frac{\left(6.626 \times 10^{-34}\ \text{J} \cdot \text{s}\right)\left(3.00 \times 10^8\ \text{m/s}\right)}{410 \times 10^{-9}\ \text{m}} = 4.85 \times 10^{-19}\ \text{J}$$

Then the sixth excited state is above the fifth by

$$(4.85 - 3.82) \times 10^{-19}\ \text{J} = 1.03 \times 10^{-19}\ \text{J}$$

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In the 6 to 5 transition the atom emits a photon with the infrared wavelength

$$\lambda = \frac{hc}{E_{65}} = \frac{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)\left(3.00\times10^{8}\ \text{m/s}\right)}{1.03\times10^{-19}\ \text{J}}$$

$$= 1.94\times10^{-6}\ \text{m} = \boxed{1.94\ \mu\text{m}}$$

P42.4 (a) Denote the energy level n of the atom by E_n. For the transition $m \to 1$, the energy of the emitted photon and its wavelength λ_{m1} are related thus:

$$\Delta E_{m1} = E_m - E_1 = \frac{hc}{\lambda_{m1}}$$

For the transition $n \to 1$, the energy of the emitted photon and its wavelength λ_{n1} are related similarly:

$$\Delta E_{n1} = E_n - E_1 = \frac{hc}{\lambda_{n1}}$$

Therefore, for the transition $m \to n$, the energy of the emitted photon and its wavelength λ_{mn} (where m is the higher state, so $\lambda_{n1} > \lambda_{m1}$) can be related as

$$\Delta E_{mn} = E_m - E_n = \frac{hc}{\lambda_{mn}}$$

$$\Delta E_{mn} = \left(E_m - E_1\right) - \left(E_n - E_1\right) = \frac{hc}{\lambda_{mn}}$$

$$= \frac{hc}{\lambda_{m1}} - \frac{hc}{\lambda_{n1}} = \frac{hc}{\lambda_{mn}} \qquad \rightarrow \qquad \frac{1}{\lambda_{mn}} = \frac{1}{\lambda_{m1}} - \frac{1}{\lambda_{n1}}$$

This result may be written as $\boxed{\lambda_{mn} = \left|\frac{1}{1/\lambda_{m1} - 1/\lambda_{n1}}\right|}$.

(b) Multiply the result of part (a) by 2π and apply the definition $k_{ij} = 2\pi/\lambda_{ij}$:

$$2\pi\left(\frac{1}{\lambda_{mn}} = \left|\frac{1}{\lambda_{m1}} - \frac{1}{\lambda_{n1}}\right|\right) \qquad \rightarrow \qquad \boxed{k_{mn} = \left|k_{m1} - k_{n1}\right|}$$

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P42.5 Our equation is $\frac{1}{\lambda} = R_{\mathrm{H}}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$ where $R_{\mathrm{H}} = 1.097\ 373\ 2 \times 10^{7}\ \mathrm{m}^{-1}$.

With our notation we have identified what Rydberg did not know, that the integers are the principal quantum numbers of the original and final atomic states in the photon emission process. For the Lyman series, we have $n_f = 1$, and $n_i = 2, 3, 4, \ldots$ We solve for the quantum number of the original state

$$\frac{1}{n_i^2} = \frac{1}{n_f^2} - \frac{1}{R_H \lambda} \quad \rightarrow \quad n_i = \left(\frac{1}{n_f^2} - \frac{1}{R_H \lambda}\right)^{-1/2}$$

(a) and substitute the given values.

$$n_i = \left(\frac{1}{1^2} - \frac{1}{94.96 \times 10^{-9}\,\mathrm{m} \times 1.097 \times 10^{7}\,\mathrm{m}^{-1}}\right)^{-1/2} = \boxed{5}$$

The electron makes a transition from energy level 5 to the ground state to emit light in this spectral line.

(b) and (c) By Figure 42.8, spectral lines in the Balmer and Paschen series all have much longer wavelengths, since much smaller energy losses put the atom into energy levels 2 or 3. The expressions

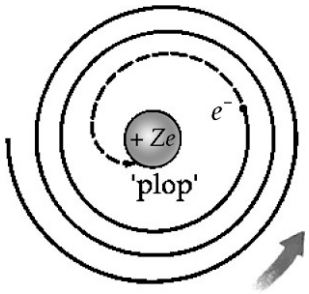
$$n_i = \left(\frac{1}{2^2} - \frac{1}{94.96 \times 10^{-9}\,\mathrm{m} \times 1.097 \times 10^{7}\,\mathrm{m}^{-1}}\right)^{-1/2}$$

and $$n_i = \left(\frac{1}{3^2} - \frac{1}{94.96 \times 10^{-9}\,\mathrm{m} \times 1.097 \times 10^{7}\,\mathrm{m}^{-1}}\right)^{-1/2}$$

are imaginary quantities, not real positive integers. The Lyman-delta wavelength given cannot be part of the Balmer or the Paschen series.

Section 42.2 Early Models of the Atom

P42.6 According to a classical model, the electron moving as a particle in uniform circular motion about the proton in the hydrogen atom experiences a force $k_e e^2/r^2$; and from Newton's second law, $F = ma$, its acceleration is $k_e e^2/m_e r^2$.



ANS. FIG. P42.6

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(a) Using the fact that the Coulomb constant is $k_e = \dfrac{1}{4\pi \in_0}$, the centripetal acceleration is

$$a = \frac{v^2}{r} = \frac{F}{m_e} = \frac{e^2}{4\pi \in_0 r^2 m_e} \qquad \rightarrow \qquad m_e v^2 = \frac{e^2}{4\pi \in_0 r}$$

The total energy is

$$E = K + U = \frac{m_e v^2}{2} - \frac{e^2}{4\pi \in_0 r} = -\frac{e^2}{8\pi \in_0 r}$$

Substitute the expressions for E and a into the relation for $\dfrac{dE}{dt}$:

$$\frac{dE}{dt} = \frac{-1}{6\pi \in_0} \frac{e^2 a^2}{c^3}$$

$$\frac{e^2}{8\pi \in_0 r^2} \frac{dr}{dt} = \frac{-e^2}{6\pi \in_0 c^3} \left(\frac{e^2}{4\pi \in_0 r^2 m_e} \right)^2$$

Therefore, $\dfrac{dr}{dt} = -\dfrac{e^4}{12\pi^2 \in_0^2 m_e^2 c^3} \left(\dfrac{1}{r^2} \right)$.

(b) From the result of part (a), we have

$$T = \int_0^T dt = -\int_{2.00\times10^{-10}\ \text{m}}^{0} \frac{12\pi^2 \in_0^2 r^2 m_e^2 c^3}{e^4} dr$$

$$= \int_0^{2.00\times10^{-10}\ \text{m}} \frac{12\pi^2 \in_0^2 r^2 m_e^2 c^3}{e^4} dr$$

$$= \frac{12\pi^2 \in_0^2 m_e^2 c^3}{e^4} \frac{r^3}{3}\bigg|_0^{2.00\times10^{-10}}$$

$$= \frac{12\pi^2 \left(8.85\times10^{-12}\ \text{C}\right)^2 \left(9.11\times10^{-31}\ \text{kg}\right)^2 \left(3.00\times10^{8}\ \text{m/s}\right)^3}{\left(1.60\times10^{-19}\ \text{C}\right)^4} \times \frac{\left(2.00\times10^{-10}\ \text{m}\right)^3}{3}$$

$$= 8.46\times10^{-10}\ \text{s} = \boxed{0.846\ \text{ns}}$$

Since atoms last much longer than 0.8 ns, the classical laws (fortunately!) do not hold for systems of atomic size.

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P42.7 (a) The point of closest approach is found when

$$E = K_i + U_i = K_f + U_f$$

$$K_i + 0 = 0 + \frac{k_e q_\alpha q_{\text{Au}}}{r_{\min}}$$

$$\rightarrow \quad r_{\min} = \frac{k_e (2e)(79e)}{K_i}$$

$$r_{\min} = \frac{\left(8.99 \times 10^9 \ \text{N} \cdot \text{m}^2/\text{C}^2\right)(158)\left(1.602 \times 10^{-19} \ \text{C}\right)^2}{\left(4.00 \times 10^6 \ \text{eV}\right)\left(1.602 \times 10^{-19} \ \text{J/eV}\right)}$$

$$= \boxed{5.69 \times 10^{-14} \ \text{m}}$$

(b) The maximum force exerted on the alpha particle is

$$F_{\max} = \frac{k_e q_\alpha q_{\text{Au}}}{r_{\min}^2} = \frac{\left(8.99 \times 10^9 \ \text{N} \cdot \text{m}^2/\text{C}^2\right)(158)\left(1.602 \times 10^{-19} \ \text{C}\right)^2}{\left(5.69 \times 10^{-14} \ \text{m}\right)^2}$$

$$= \boxed{11.3 \ \text{N}}$$

away from the nucleus.

Section 42.3 Bohr's Model of the Hydrogen Atom

***P42.8** From the equation just above Equation 42.9 in the text, $\frac{1}{2} m_e v^2 = \frac{k_e e^2}{2r}$, we have

$$v^2 = \frac{k_e e^2}{m_e r}$$

and using

$$r_n = \frac{n^2 \hbar^2}{m_e k_e e^2}$$

we obtain

$$v_n^2 = \frac{k_e e^2}{m_e \left(n^2 \hbar^2 / m_e k_e e^2\right)}$$

or

$$v_n = \frac{k_e e^2}{n\hbar}$$

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***P42.9** We use $E_n = \dfrac{-13.6\text{ eV}}{n^2}$. To ionize the atom when the electron is in the nth level, it is necessary to add an amount of energy given by

$$E = -E_n = \frac{13.6\text{ eV}}{n^2}$$

(a) Thus, in the ground state where $n = 1$, we have $\boxed{E = 13.6\text{ eV}}$.

(b) In the $n = 3$ level, $E = \dfrac{13.6\text{ eV}}{3^2} = \boxed{1.51\text{ eV}}$.

P42.10 The allowed energy levels of the hydrogen atom are given by

$$E_n = -\frac{13.6\text{ eV}}{n^2} \text{ where } n = 1, 2, 3\ldots$$

A transition in which a lower state n_i absorbs a photon of energy ΔE results in a higher state n_f, and energy is conserved:

$$E_i + \Delta E = E_f$$

or

$$\Delta E = E_f - E_i = -\frac{13.6\text{ eV}}{n_f^2} - \left(-\frac{13.6\text{ eV}}{n_i^2}\right) = 13.6\text{ eV}\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)$$

(a) For the transition $n_i = 2$ to $n_f = 5$,

$$\Delta E = 13.6\text{ eV}\left(\frac{1}{2^2} - \frac{1}{5^2}\right) = \boxed{2.86\text{ eV}}$$

(b) For the transition $n_i = 4$ to $n_f = 6$,

$$\Delta E = 13.6\text{ eV}\left(\frac{1}{4^2} - \frac{1}{6^2}\right) = \boxed{0.472\text{ eV}}$$

P42.11 The allowed energy levels of the hydrogen atom are given by

$$E_n = -\frac{13.6\text{ eV}}{n^2} \text{ where } n = 1, 2, 3\ldots$$

In a transition for higher state n_i to lower state n_f, a photon of energy ΔE is emitted, and energy is conserved:

$$E_i + \Delta E = E_f$$

or

$$\Delta E = E_f - E_i = -\frac{13.6\text{ eV}}{n_f^2} - \left(-\frac{13.6\text{ eV}}{n_i^2}\right) = 13.6\text{ eV}\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)$$

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(a) For the transition $n_i = 5$ to $n_f = 3$,

$$\Delta E = 13.6 \text{ eV}\left(\frac{1}{3^2} - \frac{1}{5^2}\right) = \boxed{0.967 \text{ eV}}$$

(b) To find the wavelength of the emitted photon, we use Equation 42.5:

$$\Delta E = 0.967 \text{ eV} = hf = \frac{hc}{\lambda}$$

Solving,

$$\lambda = \frac{hc}{\Delta E} = \frac{1\,240 \text{ eV} \cdot \text{nm}}{0.967 \text{ eV}} = 1\,282 \text{ nm} = \boxed{1.28\ \mu\text{m}}$$

(c) The frequency of the emitted photon is

$$f = \frac{c}{\lambda} = \frac{3.00 \times 10^8 \text{ m/s}}{1\,282 \times 10^{-9} \text{ m}} = \boxed{2.34 \times 10^{14} \text{ Hz}}$$

P42.12 (a) The longest wavelength implies lowest frequency and smallest energy. The electron makes a transition from $n = 3$ to $n = 2$:

$$\Delta E = -\frac{13.6 \text{ eV}}{3^2} + \frac{13.6 \text{ eV}}{2^2} = \boxed{1.89 \text{ eV}}$$

(b) The photon's wavelength is

$$\lambda = \frac{c}{f} = \frac{hc}{\Delta E} = \frac{1\,240 \text{ eV} \cdot \text{nm}}{(1.89 \text{ eV})} = \boxed{656 \text{ nm}}$$

This is the red Balmer-alpha line, which gives its characteristic color to the chromosphere of the Sun and to photographs of the Orion nebula.

(c) The shortest wavelength implies highest frequency and greatest energy. The electron makes a transition from $n = \infty$ to $n = 2$:

$$\Delta E = -\frac{13.6 \text{ eV}}{\infty} + \frac{13.6 \text{ eV}}{2^2} = \boxed{3.40 \text{ eV}}$$

(d) The photon's wavelength is

$$\lambda = \frac{c}{f} = \frac{hc}{\Delta E} = \frac{1\,240 \text{ eV} \cdot \text{nm}}{3.40 \text{ eV}} = \boxed{365 \text{ nm}}$$

(e) This is the Balmer series limit, $\boxed{365 \text{ nm}}$, in the near ultraviolet.

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P42.13 (a) From the equation just above Equation 42.9 in the text, $v_1 = \sqrt{\dfrac{k_e e^2}{m_e r_1}}$

where, from Equation 42.10,

$$r_1 = (1)^2 a_0 = 0.005\,29 \text{ nm} = 5.29 \times 10^{-11} \text{ m}$$

Substituting numerical values,

$$v_1 = \sqrt{\frac{\left(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2\right)\left(1.60 \times 10^{-19} \text{ C}\right)^2}{\left(9.11 \times 10^{-31} \text{ kg}\right)\left(5.29 \times 10^{-11} \text{ m}\right)}}$$

$$= \boxed{2.19 \times 10^6 \text{ m/s}}$$

(b) The kinetic energy of the electron is

$$K_1 = \frac{1}{2} m_e v_1^2 = \frac{1}{2}\left(9.11 \times 10^{-31} \text{ kg}\right)\left(2.19 \times 10^6 \text{ m/s}\right)^2$$

$$= 2.18 \times 10^{-18} \text{ J} = \boxed{13.6 \text{ eV}}$$

(c) The electric potential energy of the atom is

$$U_1 = -\frac{k_e e^2}{r_1} = -\frac{\left(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2\right)\left(1.60 \times 10^{-19} \text{ C}\right)^2}{5.29 \times 10^{-11} \text{ m}}$$

$$= -4.35 \times 10^{-18} \text{ J} = \boxed{-27.2 \text{ eV}}$$

***P42.14** Each atom gives up its kinetic energy in emitting a photon, so

$$\frac{1}{2} m v^2 = \frac{hc}{\lambda} = \frac{\left(6.626 \times 10^{-34} \text{ J} \cdot \text{s}\right)\left(3.00 \times 10^8 \text{ m/s}\right)}{\left(1.216 \times 10^{-7} \text{ m}\right)}$$

$$= 1.63 \times 10^{-18} \text{ J}$$

Their speed before the collision is

$$v = \sqrt{\frac{2(1.63 \times 10^{-18} \text{ J})}{1.67 \times 10^{-27} \text{ kg}}} = \boxed{4.42 \times 10^4 \text{ m/s}}$$

***P42.15** (a) The speed of the moon in its orbit is

$$v = \frac{2\pi r}{T} = \frac{2\pi\left(3.84 \times 10^8 \text{ m}\right)}{2.36 \times 10^6 \text{ s}} = 1.02 \times 10^3 \text{ m/s}$$

so,

$$L = mvr = \left(7.36 \times 10^{22} \text{ kg}\right)\left(1.02 \times 10^3 \text{ m/s}\right)\left(3.84 \times 10^8 \text{ m}\right)$$

$$= \boxed{2.89 \times 10^{34} \text{ kg} \cdot \text{m}^2/\text{s}}$$

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(b) We have $L = n\hbar$,

or $$n = \frac{L}{\hbar} = \frac{2.89 \times 10^{34}\ \text{kg} \cdot \text{m}^2/\text{s}}{1.055 \times 10^{-34}\ \text{J} \cdot \text{s}} = \boxed{2.74 \times 10^{68}}.$$

(c) We have $$n\hbar = L = mvr = m\left(\frac{GM_e}{r}\right)^{1/2} r,$$

so $$r = \frac{\hbar^2}{m^2 GM_e} n^2 = Rn^2 \text{ and } \frac{\Delta r}{r} = \frac{(n+1)^2 R - n^2 R}{n^2 R} = \frac{2n+1}{n^2},$$

which is approximately equal to $\frac{2}{n} = \boxed{7.30 \times 10^{-69}}$.

P42.16 (a) The collection of excited atoms must make these six transitions to get back to state one: $4 \to 1$, $4 \to 2$, and $4 \to 3$; $3 \to 1$ and $3 \to 2$; $2 \to 1$. Thus, the absorbed photon changes the atomic state from 1 to 4:

$$E_1 + hf = E_4 \quad \to \quad hf = E_4 - E_1, \quad \text{where} \quad E_n = -\frac{13.6\ \text{eV}}{n^2}$$

The incoming photons have energy

$$hf = \Delta E = E_f - E_i = -0.850\ \text{eV} - (-13.6\ \text{eV}) = 12.75\ \text{eV} = \frac{hc}{\lambda}$$

and wavelength

$$\lambda = \frac{hc}{\Delta E} = \frac{1\ 240\ \text{eV} \cdot \text{nm}}{12.75\ \text{eV}} = \boxed{97.3\ \text{nm}}$$

(b) The longest of the six wavelengths corresponds to the lowest photon energy, emitted in the transition $4 \to 3$. By energy conservation, $E_4 = hf + E_3$ and

$$hf = E_4 - E_3 = 13.6\ \text{eV}\left(\frac{1}{3^2} - \frac{1}{4^2}\right) = 0.661\ \text{eV} = \frac{hc}{\lambda}$$

which gives

$$\lambda = \frac{hc}{E_4 - E_3} = \frac{1\ 240\ \text{eV} \cdot \text{nm}}{0.661\ \text{eV}} = 1\ 876\ \text{nm} = \boxed{1.88\ \mu\text{m}}$$

(c) The wavelength is in the infrared region of the spectrum.

(d) The wavelength is part of the Paschen series, since the lower state has $n = 3$.

(e) The shortest wavelength emitted is from the transition $4 \to 1$, and it is the same as the wavelength absorbed: 97.3 nm.

(f) The wavelength is in the ultraviolet region of the spectrum.

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(g) The wavelength is part of the Lyman series, since the lower state has $n = 1$.

P42.17 (a) From Equation 42.12,

$$r_n = n^2 a_0 = n^2(0.052\,9\text{ nm})$$

and $r_3 = (3)^2(0.052\,9\text{ nm}) = \boxed{0.476\text{ nm}}$

(b) Using Equation 42.8, we calculate the momentum of the electron:

$$m_e v_2 = m_e\sqrt{\frac{k_e e^2}{m_e r_2}} = \sqrt{\frac{m_e k_e e^2}{r_2}}$$

$$= \sqrt{\frac{\left(9.11\times10^{-31}\text{ kg}\right)\left(8.99\times10^{9}\text{ N}\cdot\text{m}^2/\text{C}^2\right)\left(1.602\times10^{-19}\text{ C}\right)^2}{0.476\times10^{-9}\text{ m}}}$$

$$= 6.64\times10^{-25}\text{ kg}\cdot\text{m/s}$$

The de Broglie wavelength for the electron is

$$\lambda = \frac{h}{mv} = \frac{6.626\times10^{-34}\text{ J}\cdot\text{s}}{6.64\times10^{-25}\text{ kg}\cdot\text{m/s}} = 9.97\times10^{-10}\text{ m} = \boxed{0.997\text{ nm}}$$

P42.18 We note, during our calculations, that the nominal velocity of the electron is less than 1% of the speed of light; therefore, we do not need to use relativistic equations.

(a) By Bohr's theory and Equation 42.12,

$$r_n = n^2 a_0$$

$$r_2 = (2)^2\left(0.052\,9\text{ nm}\right) = \boxed{0.212\text{ nm}}$$

(b) Since $m_e v r = n\hbar$,

$$p = m_e v = \frac{n\hbar}{r} = \frac{2\left(1.054\,6\times10^{-34}\text{ J}\cdot\text{s}\right)}{2.12\times10^{-10}\text{ m}}$$

$$= \boxed{9.97\times10^{-25}\text{ kg}\cdot\text{m/s}}$$

(c) $\vec{\mathbf{L}} = \vec{\mathbf{r}}\times\vec{\mathbf{p}}$ becomes

$$L_2 = m_e v_2 r_2 = \left(9.97\times10^{-25}\text{ kg}\cdot\text{m/s}\right)\left(0.212\times10^{-9}\text{ m}\right)$$

$$= \boxed{2.11\times10^{-34}\text{ kg}\cdot\text{m}^2/\text{s}}$$

(d) Next, the speed is

$$v = \frac{p}{m_e} = \frac{9.97\times10^{-25}\text{ kg}\cdot\text{m/s}}{9.11\times10^{-31}\text{ kg}} = 1.09\times10^{6}\text{ m/s}$$

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So the kinetic energy is $K = \frac{1}{2}m_e v^2$:

$$K = \frac{\left(9.11 \times 10^{-31}\ \text{kg}\right)\left(1.09 \times 10^{6}\text{m/s}\right)^2}{2}$$

$$= \frac{5.45 \times 10^{-19}\ \text{J}}{1.602 \times 10^{-19}\ \text{J/eV}} = \boxed{3.40\ \text{eV}}$$

(e) From Chapter 25, the electric potential energy is $U = k_e \frac{q_1 q_2}{r}$:

$$U = -\frac{k_e e^2}{r} = -\frac{\left(8.99 \times 10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left(1.602 \times 10^{-19}\ \text{C}\right)^2}{2.12 \times 10^{-10}\ \text{m}}$$

$$= -1.09 \times 10^{-18}\ \text{J} = \boxed{-6.80\ \text{eV}}$$

(f) Thus the total energy is

$$E = K + U = -5.45 \times 10^{-19}\ \text{J} = \boxed{-3.40\ \text{J}}$$

P42.19 (a) The photon has energy 2.28 eV, and $\frac{13.6\ \text{eV}}{2^2} = 3.40\ \text{eV}$ is required to ionize a hydrogen atom from state $n = 2$. So while the photon cannot ionize a hydrogen atom pre-excited to $n = 2$, it can ionize a hydrogen atom in the $n = \boxed{3}$ state, with energy $-\frac{13.6\ \text{eV}}{3^2} = -1.51\ \text{eV}$.

(b) The electron thus freed can have kinetic energy

$$K_e = 2.28\ \text{eV} - 1.51\ \text{eV} = 0.769\ \text{eV} = \frac{1}{2}m_e v^2$$

Therefore,

$$v = \sqrt{\frac{2(0.769\ \text{eV})\left(1.60\times10^{-19}\ \text{J/eV}\right)}{9.11\times10^{-31}\ \text{kg}}} = 5.20\times10^{5}\ \text{m/s}$$

$$= \boxed{520\ \text{km/s}}$$

P42.20 (a) From the Bohr theory, we find the speed of the electron:

$$L = m_e v r = n\hbar \qquad \rightarrow \qquad v = \frac{n\hbar}{m_e r}$$

The period of its orbital motion is $T = \frac{2\pi r}{v} = \frac{2\pi r m_e r}{n\hbar}$.

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Substituting the orbital radius $r = \dfrac{n^2\hbar^2}{m_e k_e e^2}$, we find

$$T = \frac{2\pi m_e n^4 \hbar^4}{n\hbar m_e^2 k_e^2 e^4} = \frac{2\pi\hbar^3}{m_e k_e^2 e^4} n^3$$

Thus we have the periods determined in terms of the ground-state period

$$t_0 = \frac{2\pi\hbar^3}{m_e k_e^2 e^4}$$

$$= \frac{2\pi(1.054\ 6\times10^{-34}\ \text{J}\cdot\text{s})^3}{(9.11\times10^{-31}\ \text{kg})(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2)^2(1.602\times10^{-19}\ \text{C})^4}$$

$$= 1.52\times10^{-16}\ \text{s} = 152\times10^{-18}\ \text{s} = \boxed{152\ \text{as}}$$

(b) In the $n = 2$ state, the period is

$$T = t_0 n^3 = t_0(2)^3 = 8t_0 = 1.22\times10^{-15}\ \text{s}$$

The number of orbits completed in the excited state is

$$N = \frac{10\times10^{-6}\ \text{s}}{T} = \frac{10\times10^{-6}\ \text{s}}{1.22\times10^{-15}\ \text{s}} = \boxed{8.23\times10^{9}\ \text{revolutions}}$$

(c) Its lifetime in electron years is comparable to the lifetime of the Sun in Earth years, so we can think of it as a long time.

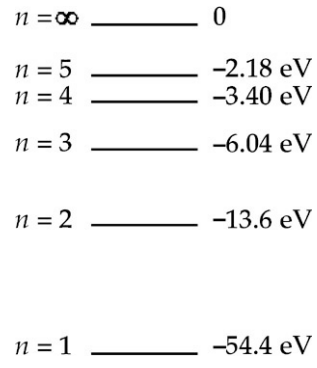
P42.21 (a) The energy levels of a hydrogen-like ion whose charge number is Z are given by

$$E_n = (-13.6\ \text{eV})\frac{Z^2}{n^2}$$

Thus for helium ($Z = 2$), the energy levels are

$$\boxed{E_n = -\frac{54.4\ \text{eV}}{n^2} \quad n = 1,\ 2,\ 3,\ \ldots}$$

The enegy level diagram for helium is shown in ANS. FIG. P42.21.



ANS. FIG. P42.21

(b) For He^+, $Z = 2$, so we see that the ionization energy (the energy required to take the electron from the $n = 1$ to the $n = \infty$ state) is

$$E = E_\infty - E_1 = 0 - \frac{(-13.6\ \text{eV})(2)^2}{(1)^2} = \boxed{54.4\ \text{eV}}$$

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Section 42.4 The Quantum Model of the Hydrogen Atom

P42.22 The reduced mass of positronium is **less** than hydrogen, so the photon energy will be **less** for positronium than for hydrogen. This means that the wavelength of the emitted photon will be **longer** than 656.3 nm. On the other hand, helium has about the same reduced mass but more charge than hydrogen, so its transition energy will be **larger**, corresponding to a wavelength **shorter** than 656.3 nm.

All the factors in the given equation are constant for this problem except for the reduced mass and the nuclear charge. Therefore, the wavelength corresponding to the energy difference for the transition can be found simply from the ratio of mass and charge variables.

For hydrogen, $\mu = \dfrac{m_p m_e}{m_p + m_e} \approx m_e$. The photon energy is $\Delta E = E_3 - E_2$.

Its wavelength is $\lambda = 656.3$ nm, where $\lambda = \dfrac{c}{f} = \dfrac{hc}{\Delta E}$.

(a) For positronium, $\mu = \dfrac{m_e m_e}{m_e + m_e} = \dfrac{m_e}{2}$,

so the energy of each level is one half as large as in hydrogen. The photon energy is inversely proportional to its wavelength, so for positronium,

$$\lambda_{32} = 2(656.3 \text{ nm}) = \boxed{1.31\ \mu\text{m}} \quad \text{(in the infrared region)}$$

(b) For He^+, $\mu \approx m_e$, $q_1 = e$, and $q_2 = 2e$, so the transition energy is $2^2 = 4$ times larger than hydrogen. Then,

$$\lambda_{32} = \left(\frac{656}{4}\right) \text{nm} = \boxed{164 \text{ nm}} \quad \text{(in the ultraviolet region)}$$

P42.23 (a) For this problem, refer to the equation from Problem 22, with $q_1 = q_2 = e$. For a particular transition from n_i to n_f,

$$\Delta E_{\text{H}} = -\frac{\mu_{\text{H}} k_e^{\,2} e^4}{2\hbar^2}\left(\frac{1}{n_f^{\,2}} - \frac{1}{n_i^{\,2}}\right) = \frac{hc}{\lambda_{\text{H}}}$$

and $$\Delta E_{\text{D}} = -\frac{\mu_{\text{D}} k_e^{\,2} e^4}{2\hbar^2}\left(\frac{1}{n_f^{\,2}} - \frac{1}{n_i^{\,2}}\right) = \frac{hc}{\lambda_{\text{D}}},$$

where $\mu_{\text{H}} = \dfrac{m_e m_p}{m_e + m_p}$ and $\mu_{\text{D}} = \dfrac{m_e m_{\text{D}}}{m_e + m_{\text{D}}}$.

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By division, $\frac{\Delta E_{\mathrm{H}}}{\Delta E_{\mathrm{D}}} = \frac{\mu_{\mathrm{H}}}{\mu_{\mathrm{D}}} = \frac{\lambda_{\mathrm{D}}}{\lambda_{\mathrm{H}}}$ or $\lambda_{\mathrm{D}} = \left(\frac{\mu_{\mathrm{H}}}{\mu_{\mathrm{D}}}\right)\lambda_{\mathrm{H}}$. Then,

$$\boxed{\lambda_{\mathrm{H}} - \lambda_{\mathrm{D}} = \left(1 - \frac{\mu_{\mathrm{H}}}{\mu_{\mathrm{D}}}\right)\lambda_{\mathrm{H}}}$$

(b) $$\frac{\mu_{\mathrm{H}}}{\mu_{\mathrm{D}}} = \left(\frac{m_e m_p}{m_e + m_p}\right)\left(\frac{m_e + m_{\mathrm{D}}}{m_e m_{\mathrm{D}}}\right)$$

$$= \frac{(1.007\ 276\ \mathrm{u})(0.000\ 549\ \mathrm{u} + 2.013\ 553\ \mathrm{u})}{(0.000\ 549\ \mathrm{u} + 1.007\ 276\ \mathrm{u})(2.013\ 553\ \mathrm{u})}$$

$$= 0.999\ 728$$

$$\lambda_{\mathrm{H}} - \lambda_{\mathrm{D}} = (1 - 0.999\ 728)(656.3\ \mathrm{nm}) = \boxed{0.179\ \mathrm{nm}}$$

P42.24 (a) The uncertainty principle is represented by $\Delta x \Delta p \geq \frac{\hbar}{2}$.

Thus, if $\Delta x = r$, $\boxed{\Delta p \geq \frac{\hbar}{2r}}$.

(b) The minimum uncertainty would be attained only if the wave function had a particular (gaussian) waveform. We assume that the momentum uncertainty is just twice as large as its minimum possible value: $\Delta p = \frac{\hbar}{r}$. Then the kinetic energy is

$$K = \frac{p^2}{2m_e} \approx \frac{(\Delta p)^2}{2m_e} = \boxed{\frac{\hbar^2}{2m_e r^2}}$$

(c) The electric potential energy is $U = -\frac{k_e e^2}{r}$ so the total energy is

$$E = K + U \approx \boxed{\frac{\hbar^2}{2m_e r^2} - \frac{k_e e^2}{r}}.$$

(d) To minimize E as a function of r, we require

$$\frac{dE}{dr} = -\frac{\hbar^2}{m_e r^3} + \frac{k_e e^2}{r^2} = 0 \quad \rightarrow \quad r = \boxed{\frac{\hbar^2}{m_e k_e e^2} = a_0} \text{ (the Bohr radius)}$$

(e) Then the energy is

$$E = \frac{\hbar^2}{2m_e}\left(\frac{m_e k_e e^2}{\hbar^2}\right)^2 - k_e e^2\left(\frac{m_e k_e e^2}{\hbar^2}\right) = -\frac{m_e k_e^2 e^4}{2\hbar^2}$$

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Substituting numerical values,

$$E = -\frac{m_e k_e^2 e^4}{2\hbar^2} = -\frac{m_e k_e^2 e^4}{2\hbar^2}$$

$$= -\frac{\left(9.109\times10^{-31}\ \text{kg}\right)\left(8.988\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)^2\left(1.602\times10^{-19}\ \text{C}\right)^4}{2\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}/2\pi\right)^2}$$

$$= -2.179\times10^{-18}\ \text{J}\left(\frac{1\ \text{eV}}{1.602\times10^{-19}\ \text{J}}\right)$$

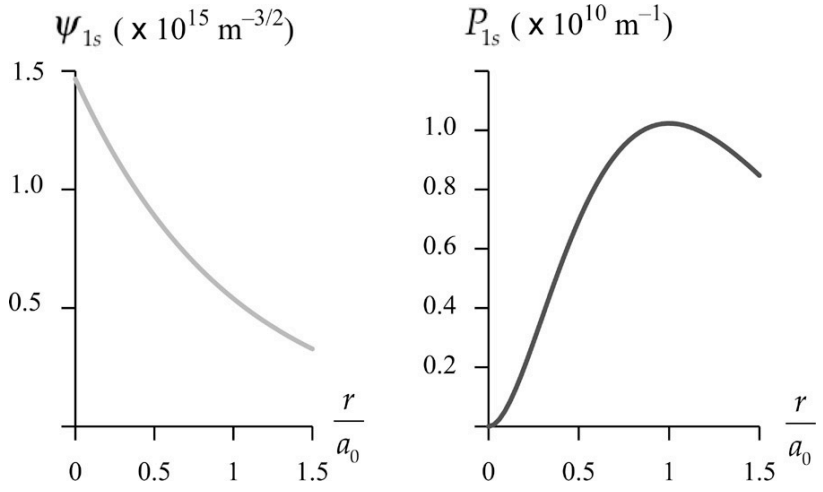
$$= \boxed{-13.6\ \text{eV}}$$

(f) With our particular choice for the momentum uncertainty as double its minimum possible value, we find our results are in agreement with the Bohr theory.

Section 42.5 The Wave Functions for Hydrogen

P42.25 $\psi_{1s}(r) = \dfrac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}$ is the ground state hydrogen wave function.

$P_{1s}(r) = \dfrac{4r^2}{a_0^3} e^{-2r/a_0}$ is the ground state radial probability distribution function. The plots are shown in ANS. FIG. P42.25.



ANS. FIG. P42.25

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P42.26 (a) We first find the first and second derivatives of the wave function:

$$\psi = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0} \quad \rightarrow \quad \frac{2}{r}\frac{d\psi}{dr} = \frac{2}{r}\left(\frac{-1}{\sqrt{\pi a_0^5}} e^{-r/a_0} \right) = -\frac{2}{r a_0}\psi$$

and $$\frac{d^2\psi}{dr^2} = \frac{1}{\sqrt{\pi a_0^7}} e^{-r/a_0} = \frac{1}{a_0^2}\psi$$

Substitution into the Schrödinger equation to test the validity of the solution yields

$$-\frac{\hbar^2}{2m_e}\left(\frac{1}{a_0^2} - \frac{2}{r a_0} \right)\psi - \frac{e^2}{4\pi \in_0 r}\psi = E\psi$$

But, from Equation 42.11, $a_0 = \dfrac{\hbar^2}{m_e k_e e^2} = \dfrac{\hbar^2 \left(4\pi \in_0\right)}{m_e e^2}$, thus

$$-\frac{\hbar^2}{2m_e}\left(\frac{1}{a_0^2} - \frac{2}{r a_0} \right)\psi - \frac{e^2}{4\pi \in_0 r}\psi = E\psi$$

$$-\frac{\hbar^2}{2m_e}\frac{1}{a_0^2}\psi + \left[\frac{\hbar^2}{m_e r}\frac{1}{a_0}\psi - \frac{e^2}{4\pi \in_0 r}\psi \right] = E\psi$$

$$-\frac{\hbar^2}{2m_e}\frac{1}{a_0^2}\psi + \left[\frac{\cancel{\hbar^2}}{\cancel{m_e} r}\frac{\cancel{m_e} e^2}{4\pi \in_0 \cancel{\hbar^2}} - \frac{e^2}{4\pi \in_0 r} \right]\psi = E\psi$$

$$-\frac{\hbar^2}{2m_e}\frac{1}{a_0^2}\psi = E\psi$$

The Schrödinger equation is satisfied if $E = -\dfrac{\hbar^2}{2m_e}\dfrac{1}{a_0^2}$.

(b) Substituting $a_0 = \dfrac{\hbar^2}{m_e k_e e^2}$ for one factor of a_0, we find that

$$E = -\frac{\hbar^2}{2m_e}\frac{1}{a_0^2} = -\frac{\cancel{\hbar^2}}{2\cancel{m_e}}\frac{1}{a_0}\frac{\cancel{m_e} k_e e^2}{\cancel{\hbar^2}} = \boxed{E = -\frac{k_e e^2}{2a_0}}$$

P42.27 The wave function given is

$$\psi = \frac{1}{\sqrt{3}}\frac{1}{\left(2a_0\right)^{3/2}}\frac{r}{a_0}e^{-r/2a_0}$$

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so, by Equation 42.24,

$$P_r = 4\pi r^2 |\psi|^2 = 4\pi r^2 \frac{r^2}{24a_0^5} e^{-r/a_0}$$

Setting $\frac{dP}{dr} = \frac{4\pi}{24a_0^5}\left[4r^3 e^{-r/a_0} + r^4\left(-\frac{1}{a_0}\right)e^{-r/a_0}\right] = 0$, we obtain

$$\left[4r^3 + -\frac{r^4}{a_0}\right] = 0$$

Solving for r, this is a maximum at $\boxed{r = 4a_0}$.

P42.28 (a) $\int |\psi|^2 dV = 4\pi \int_0^\infty |\psi|^2 r^2 dr = 4\pi\left(\frac{1}{\pi a_0^3}\right)\int_0^\infty r^2 e^{-2r/a_0} dr$. Using integral tables,

$$\int |\psi|^2 dV = -\frac{2}{a_0^2}\left[e^{-2r/a_0}\left(r^2 + a_0 r + \frac{a_0^2}{2}\right)\right]_0^\infty = \left(-\frac{2}{a_0^2}\right)\left(-\frac{a_0^2}{2}\right) = \boxed{1}$$

so the wave function as given is normalized.

(b) $P_{a_0/2 \to 3a_0/2} = 4\pi \int_{a_0/2}^{3a_0/2} |\psi|^2 r^2 dr = 4\pi\left(\frac{1}{\pi a_0^3}\right)\int_{a_0/2}^{3a_0/2} r^2 e^{-2r/a_0} dr$. Again, using integral tables,

$$P_{a_0/2 \to 3a_0/2} = -\frac{2}{a_0^2}\left[e^{-2r/a_0}\left(r^2 + a_0 r + \frac{a_0^2}{2}\right)\right]_{a_0/2}^{3a_0/2}$$

$$= -\frac{2}{a_0^2}\left[e^{-3}\left(\frac{17a_0^2}{4}\right) - e^{-1}\left(\frac{5a_0^2}{4}\right)\right] = \boxed{0.497}$$

P42.29 The hydrogen ground-state radial probability density is, from Equation 42.25,

$$P_{1s}(r) = 4\pi r^2 |\psi_{1s}|^2 = \frac{4r^2}{a_0^3}\exp\left(-\frac{2r}{a_0}\right)$$

The number of observations at $2a_0$ is, by proportion,

$$N = (1\ 000)\frac{P_{1s}(2a_0)}{P_{1s}(a_0/2)} = (1\ 000)\frac{(2a_0)^2}{(a_0/2)^2}\frac{e^{-4a_0/a_0}}{e^{-a_0/a_0}} = (1\ 000)(16)e^{-3}$$

$$= \boxed{797 \text{ times}}$$

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Section 42.6 Physical Interpretation of the Quantum Numbers

P42.30 (a) In the 3*d* subshell, $n = 3$ and $\ell = 2$, we have

n	3	3	3	3	3	3	3	3	3	3
ℓ	2	2	2	2	2	2	2	2	2	2
m_ℓ	+2	+2	+1	+1	0	0	–1	–1	–2	–2
m_s	+1/2	–1/2	+1/2	–1/2	+1/2	–1/2	+1/2	–1/2	+1/2	–1/2

(a total of 10 states.)

(b) In the 3*p* subshell, $n = 3$ and $\ell = 1$, we have

n	3	3	3	3	3	3
ℓ	1	1	1	1	1	1
m_ℓ	+1	+1	0	0	–1	–1
m_s	+1/2	–1/2	+1/2	–1/2	+1/2	–1/2

(a total of 6 states.)

***P42.31** From Equation 42.27, $L = \sqrt{\ell(\ell+1)}\hbar$ (suppressing units):

$$4.714\times10^{-34} = \sqrt{\ell(\ell+1)}\left(\frac{6.626\times10^{-34}}{2\pi}\right)$$

$$\ell(\ell+1) = \frac{\left(4.714\times10^{-34}\right)^2(2\pi)^2}{\left(6.626\times10^{-34}\right)^2} = 1.998\times10^{1} \approx 20 = 4(4+1)$$

so $\boxed{\ell = 4}$.

P42.32 (a) For a 3*d* state, $n = 3$ and $\ell = 2$. Therefore,

$$L = \sqrt{\ell(\ell+1)}\hbar = \boxed{\sqrt{6}\hbar} = 2.58\times10^{-34}\ \text{J}\cdot\text{s}$$

(b) m_ℓ can have the values –2, –1, 0, 1, and 2,

so $\boxed{L_z \text{ can have the values } -2\hbar,\ -\hbar,\ 0,\ \hbar \text{ and } 2\hbar}$.

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(c) Using the relation $\cos\theta = \frac{L_z}{L}$, we find the possible values of θ:

$$\boxed{145°,\ 114°,\ 90.0°,\ 65.9°, \text{ and } 35.3°}$$

P42.33 From Equation 42.27, $L = \sqrt{\ell(\ell+1)}\hbar$.

(a) For the *d* state, $\ell = 2,$ and $L = \boxed{\sqrt{6}\hbar = 2.58 \times 10^{-34}\ \text{J}\cdot\text{s}}$.

(b) For the *f* state, $\ell = 3,$ and $L = \sqrt{12}\hbar = \boxed{2\sqrt{3}\hbar = 3.65 \times 10^{-34}\ \text{J}\cdot\text{s}}$.

P42.34 (a) For $n = 1$, we have $\ell = 0,\ m_\ell = 0,\ m_s = \pm\frac{1}{2}$.

n	ℓ	m_ℓ	m_s
1	0	0	–1/2
1	0	0	+1/2

This yields $2n^2 = 2(1)^2 = \boxed{2}$ sets.

(b) For $n = 2$, we have

n	ℓ	m_ℓ	m_s
2	0	0	±1/2
2	1	–1	±1/2
2	1	0	±1/2
2	1	+1	±1/2

This yields $2n^2 = 2(2)^2 = \boxed{8}$ sets.

Note that the number is twice the number of m_ℓ values. Also, for each ℓ there are $(2\ell+1)$ different m_ℓ values. Finally, l can take on values ranging from 0 to $n-1$.

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So the general expression is number $= \sum_{\ell=0}^{n-1} 2(2\ell+1)$.

The series is an arithmetic progression like 2 + 6 + 10 + 14.

The sum is $\sum_{0}^{n-1} 4\ell + \sum_{0}^{n-1} 2 = 4\left[\frac{n^2 - n}{2}\right] + 2n = 2n^2$.

(c) $n = 3$: $2(1) + 2(3) + 2(5) = 2 + 6 + 10 = 18$ or $2n^2 = 2(3)^2 = \boxed{18}$

(d) $n = 4$: $2(1) + 2(3) + 2(5) + 2(7) = 32$ or $2n^2 = 2(4)^2 = \boxed{32}$

(e) $n = 5$: $32 + 2(9) = 32 + 18 = 50$ or $2n^2 = 2(5)^2 = \boxed{50}$

P42.35 In the N shell, $n = 4$. For $n = 4$, ℓ can take on values of 0, 1, 2, and 3. For each value of ℓ, m_ℓ can be $-\ell$ to ℓ in integral steps. Thus, the maximum value for m_ℓ is 3. Since $L_z = m_\ell \hbar$, the maximum value for L_z is $L_z = \boxed{3\hbar}$.

P42.36 (a) Modeling it as a solid sphere, the density of a proton is,

$$\rho = \frac{m}{V} = \frac{1.67 \times 10^{-27}\ \text{kg}}{(4/3)\pi\left(1.00 \times 10^{-15}\ \text{m}\right)^3} = \boxed{3.99 \times 10^{17}\ \text{kg/m}^3}$$

(b) The radius of an electron modelled as a solid sphere is,

$$r = \left(\frac{3m}{4\pi\rho}\right)^{1/3} = \left[\frac{3\left(9.11 \times 10^{-31}\ \text{kg}\right)}{4\pi\left(3.99 \times 10^{17}\ \text{kg/m}^3\right)}\right]^{1/3}$$
$$= 8.17 \times 10^{-17}\ \text{m} = 81.7 \times 10^{-18}\ \text{m} = \boxed{8.17\ \text{am}}$$

(c) The moment of inertia of the spinning electron is

$$I = \frac{2}{5}mr^2 = \frac{2}{5}\left(9.11 \times 10^{-31}\ \text{kg}\right)\left(8.17 \times 10^{-17}\ \text{m}\right)^2$$
$$= 2.43 \times 10^{-63}\ \text{kg} \cdot \text{m}^2$$

$$L_z = I\omega = \frac{\hbar}{2} = \frac{Iv}{r}$$

Therefore,

$$v = \frac{\hbar r}{2I} = \frac{\left(6.626 \times 10^{-34}\ \text{J} \cdot \text{s}\right)\left(8.17 \times 10^{-17}\ \text{m}\right)}{2\pi\left(2 \times 2.43 \times 10^{-63}\ \text{kg} \cdot \text{m}^2\right)}$$
$$= 1.77 \times 10^{12}\ \text{m/s} = \boxed{1.77\ \text{Tm/s}}$$

(d) It is $5.91 \times 10^3 c$, which is huge compared with the speed of light and impossible.

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P42.37 The 5th excited state has $n = 6$, energy $E_6 = \dfrac{-13.6 \text{ eV}}{(6)^2} = -0.378 \text{ eV}$.

The atom loses this much energy:

$$E = \frac{hc}{\lambda} = \frac{\left(6.626 \times 10^{-34} \text{ J} \cdot \text{s}\right)\left(3.00 \times 10^{8} \text{ m/s}\right)}{\left(1\,090 \times 10^{-9} \text{ m}\right)\left(1.60 \times 10^{-19} \text{ J/eV}\right)} = 1.14 \text{ eV}$$

to end up with energy $-0.378 \text{ eV} - 1.14 \text{ eV} = -1.52 \text{ eV}$

which is the energy in state 3: $-\dfrac{13.6 \text{ eV}}{3^3} = -1.51 \text{ eV}$

While $n = 3$, ℓ can be as large as 2, giving angular momentum

$$L = \sqrt{\ell(\ell+1)}\hbar = \boxed{\sqrt{6}\hbar = 2.58 \times 10^{-34} \text{ J} \cdot \text{s}}$$

P42.38 The energy of the photon is

$$E_{ph} = \frac{1\,240 \text{ eV} \cdot \text{nm}}{88.0 \text{ eV}} = 14.1 \text{ eV}$$

The maximum energy of the ejected photoelectron from the aluminum surface is

$$K_{\max} = E_{ph} - \phi = 14.1 \text{ eV} - 4.08 \text{ eV} = 10.0 \text{ eV}$$

where the work function ϕ for aluminum is found from Table 40.1. This electron energy is not enough to excite the hydrogen atom from its ground state to even the first exited state.

P42.39 The 3*d* subshell has $n = 3$ and $\ell = 2$. Also, we have $s = 1$. Altogether we can have $n = 3$, $\ell = 2$, $m_\ell = -2, -1, 0, 1, 2$, $s = 1$, and $m_s = -1, 0, 1$, leading to the following table:

n	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
ℓ	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
m_ℓ	–2	–2	–2	–1	–1	–1	0	0	0	1	1	1	2	2	2
s	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
m_s	–1	0	1	–1	0	1	–1	0	1	–1	0	1	–1	0	1

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Section 42.7 The Exclusion Principle and the Periodic Table

P42.40 (a) The 4s subshell, for potassium and calcium, before the 3*d* subshell starts to fill for scandium through zinc.

(b) We would expect $[\text{Ar}]3d^4 4s^2$ to have lower energy, but $[\text{Ar}]3d^5 4s^1$ has more unpaired spins and lower energy according to Hund's rule.

(c) It is the ground-state configuration of chromium.

P42.41 (a) $1s^2 2s^2 2p^4$

(b) For the 1*s* electrons,

$$n = 1,\ \ell = 0,\ m_\ell = 0,\ m_s = +\frac{1}{2} \text{ and } -\frac{1}{2}$$

For the two 2*s* electrons,

$$n = 2,\ \ell = 0,\ m_\ell = 0,\ m_s = +\frac{1}{2} \text{ and } -\frac{1}{2}$$

For the four 2*p* electrons,

$$n = 2,\ \ell = 1,\ m_\ell = -1,\ 0,\ 1, \text{ and } m_s = +\frac{1}{2} \text{ and } -\frac{1}{2}$$

P42.42 Electronic configuration: sodium to argon

Orbitals 1*s*, 2*s*, and 2*p* are filled (and not shown).

	3*s*		3*p*		4*s*	
Na^{11}	⇑					$[1s^2 2s^2 2p^6]3s^1$
Mg^{12}	⇑⇓					$[1s^2 2s^2 2p^6]3s^2$
Al^{13}	⇑⇓	⇑				$[1s^2 2s^2 2p^6]3s^2 3p^1$
Si^{14}	⇑⇓	⇑	⇑			$[1s^2 2s^2 2p^6]3s^2 3p^2$
P^{15}	⇑⇓	⇑	⇑	⇑		$[1s^2 2s^2 2p^6]3s^2 3p^3$
S^{16}	⇑⇓	⇑⇓	⇑	⇑		$[1s^2 2s^2 2p^6]3s^2 3p^4$
Cl^{17}	⇑⇓	⇑⇓	⇑⇓	⇑		$[1s^2 2s^2 2p^6]3s^2 3p^5$
Ar^{18}	⇑⇓	⇑⇓	⇑⇓	⇑⇓		$[1s^2 2s^2 2p^6]3s^2 3p^6$
K^{19}	⇑⇓	⇑⇓	⇑⇓	⇑⇓	⇑	$[1s^2 2s^2 2p^6 3s^2 3p^6]4s^1$

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P42.43 In the table of electronic configurations in the text, or on a periodic table, we look for the element whose last electron is in a $3p$ state and which has three electrons outside a closed shell. Its electron configuration then ends in $3s^2\,3p^1$. The element is aluminum.

P42.44 (a) Note that the possible values for ℓ range from zero to $n - 1$.

$n + \ell$	1	2	3	4	5	6	7
subshell	1*s*	2*s*	2*p*, 3*s*	3*p*, 4*s*	3*d*, 4*p*, 5*s*	4*d*, 5*p*, 6*s*	4*f*, 5*d*, 6*p*, 7*s*

The order is 1*s*, 2*s*, 2*p*, 3*s*, 3*p*, 4*s*, 3*d*, 4*p*, 5*s*, 4*d*, 5*p*, 6*s*, 4*f*, 5*d*, 6*p*, 7*s*.

P42.45 (a) For electron one and also for electron two, $n = 3$ and $\ell = 1$; possible values are $m_\ell = 1, 0, -1$ and $m_s = 1/2, -1/2$. The exclusion principle requires that the electrons cannot have identical sets of quantum numbers. The possible states are listed here in columns giving the other quantum numbers:

electron	m_ℓ	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0
one	m_s	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
electron	m_ℓ	1	0	0	–1	–1	1	0	0	–1	–1	1	1	0	–1	–1
two	m_s	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$

electron	m_ℓ	0	0	0	0	0	–1	–1	–1	–1	–1	–1	–1	–1	–1	–1
one	m_s	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$
electron	m_ℓ	1	1	0	–1	–1	1	1	0	0	–1	1	1	0	0	–1
two	m_s	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$

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There are $6\times 5=\boxed{30}$ allowed states, since electron one can have any of three possible values for m_ℓ for both spin up and spin down, amounting to six states, and the second electron can have any of the other five states.

(b) Were it not for the exclusion principle, there would be $6\times 6=\boxed{36}$ possible states, six for each electron independently.

P42.46 Listing subshells in the order of filling, we have for element 110,

$$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^6 6s^2 4f^{14} 5d^{10} 6p^6 7s^2 5f^{14} 6d^8$$

In order of increasing principal quantum number, this is

$$\boxed{1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 4f^{14} 5s^2 5p^6 5d^{10} 5f^{14} 6s^2 6p^6 6d^8 7s^2}$$

P42.47 In the ground state of sodium, the outermost electron is in an *s* state. This state is spherically symmetric, so it generates no magnetic field by orbital motion, and has the same energy no matter whether the electron is spin-up or spin-down. The energies of the states $3p\uparrow$ and $3p\downarrow$ above 3*s* are $hf_1=\dfrac{hc}{\lambda}$ and $hf_2=\dfrac{hc}{\lambda_2}$.

The energy difference is

$$2\mu_B B = hc\left(\frac{1}{\lambda_1}-\frac{1}{\lambda_2}\right)$$

so

$$\begin{aligned} B &= \frac{hc}{2\mu_B}\left(\frac{1}{\lambda_1}-\frac{1}{\lambda_2}\right) \\ &= \frac{\left(6.626\times 10^{-34}\ \text{J}\cdot\text{s}\right)\left(3.00\times 10^{8}\ \text{m/s}\right)}{2\left(9.27\times 10^{-24}\ \text{J/T}\right)} \\ &\qquad \times\left(\frac{1}{588.995\times 10^{-9}\ \text{m}}-\frac{1}{589.592\times 10^{-9}\ \text{m}}\right) \\ &= \boxed{18.4\ \text{T}} \end{aligned}$$

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Section 42.8 More on Atomic Spectra: Visible and X-Ray

***P42.48** Some electrons can give all their kinetic energy $K_e = e\Delta V$ to the creation of a single photon of x-radiation, with

$$hf = \frac{hc}{\lambda} = e\Delta V$$

$$\lambda = \frac{hc}{e\Delta V} = \frac{(6.626\ 1\times10^{-34}\ \text{J}\cdot\text{s})(2.997\ 9\times10^{8}\ \text{m/s})}{(1.602\ 2\times10^{-19}\ \text{C})\Delta V}$$

$$= \boxed{\frac{1\ 240\ \text{nm}\cdot\text{V}}{\Delta V}}$$

P42.49 A photon of maximum energy or minimum wavelength is produced when the electron gives up all of its kinetic energy in a single collision within the target. Thus,

$$E_{\max} = \frac{hc}{\lambda_{\min}} = KE = e\Delta V$$

For a minimum wavelength of $\lambda_{\min} = 70.0\ \text{pm} = 70.0\times10^{-12}\ \text{m}$, the required accelerating voltage is

$$\Delta V = \frac{hc}{e\lambda_{\min}} = \frac{(6.626\times10^{-34}\ \text{J}\cdot\text{s})(3.00\times10^{8}\ \text{m/s})}{(1.60\times10^{-19}\ \text{C})(70.0\times10^{-12}\ \text{m})}$$

$$= 1.77\times10^{4}\ \text{V} = \boxed{17.7\ \text{kV}}$$

P42.50 The shortest wavelength is produced when the electron gives up all of its kinetic energy as a photon in a single collision within the target. For an accelerating voltage of 40.0 keV, the kinetic energy of the electrons is

$$KE = e\Delta V = e(40.0\ \text{kV}) = 40.0\ \text{keV}$$

For the shortest wavelength produced,

$$E_{\max} = \frac{hc}{\lambda_{\min}} = KE = e\Delta V$$

and $$\lambda_{\min} = \frac{hc}{e\Delta V} = \frac{1\ 240\ \text{eV}\cdot\text{nm}}{(40.0\times10^{3}\ \text{eV})} = \boxed{0.031\ 0\ \text{nm}}$$

P42.51 (a) For bismuth, $Z = 83$. Following Example 42.5, the electron in the M shell ($n = 3$) is shielded from the nuclear charge by one electron in the L shell ($n = 1$) and eight electrons in the K shell ($n = 2$). Its energy is

$$E_{\text{M}} \approx -(Z-9)^2\frac{13.6\ \text{eV}}{(3)^2} = -13.6\ \text{eV}\frac{(74)^2}{(3)^2}$$

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The electrons in the L shell ($n = 2$) are shielded from the nuclear charge by one electron in the K shell, so (from page 1324)

$$E_L \approx -(Z-1)^2 \frac{13.6 \text{ eV}}{(2)^2} = -13.6 \text{ eV} \frac{(82)^2}{(2)^2}$$

When the electron drops from the M to the L shell of the atom, it emits a photon of energy

$$E_{photon} = E_M - E_L \approx 13.6 \text{ eV}\left[-\frac{(74)^2}{(3)^2} + \frac{(82)^2}{(2)^2}\right]$$

$$= 1.46 \times 10^4 \text{ eV} \approx \boxed{15 \text{ keV}}$$

(b) The wavelength of the emitted x-ray is given by

$$\lambda = \frac{1.240 \text{ keV} \cdot \text{nm}}{E} = \frac{1.240 \text{ keV} \cdot \text{nm}}{15 \text{ keV}}$$

$$\approx 0.083 \text{ nm} = \boxed{8.3 \times 10^{-11} \text{ m}}$$

P42.52 (a) For the $3p$ state, $E_n = \dfrac{-13.6 \text{ eV } Z_{eff}^2}{n^2}$ becomes

$$-3.0 \text{ eV} = \frac{-13.6 \text{ eV } Z_{eff}^2}{3^2} \quad \text{so} \quad Z_{eff} = \boxed{1.4}$$

For the $3d$ state

$$-1.5 \text{ eV} = \frac{-13.6 \text{ eV } Z_{eff}^2}{3^2} \quad \text{so} \quad Z_{eff} = \boxed{1.0}$$

(b) When the outermost electron in sodium is promoted from the $3s$ state into a $3p$ state, its wave function still overlaps somewhat with the ten electrons below it. It therefore sees the $+11e$ nuclear charge not fully screened, and on the average moves in an electric field like that created by a particle with charge $+11e - 9.6e = 1.4e$. When this valence electron is lifted farther to a $3d$ state, it is essentially entirely outside the cloud of ten electrons below it, and moves in the field of a net charge $+11e - 10e = 1e$.

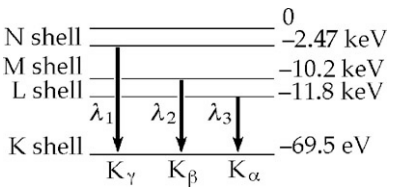
P42.53 (a) Recall $\ell \le n - 1$. For $n = 3$, $\ell = 0, 1, 2$. $\boxed{\text{If } \ell = 2, \text{ then } m_\ell = 2, 1, 0, -1, -2; \text{ if } \ell = 1, \text{ then } m_\ell = 1, 0, -1; \text{ if } \ell = 0, \text{ then } m_\ell = 0}$.

(b) The He^+ ion is a one-electron atom, so all states are have the same energy, determined by the principal quantum number n:

$$E_3 = -\frac{Z^2 E_0}{n^2} = -\frac{2^2 (13.606 \text{ eV})}{3^2} = \boxed{-6.05 \text{ eV}}$$

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***P42.54** The K series includes transitions from higher levels down to the K shell ($n = 1$). Transitions from higher n produce photons of higher energy. The ionization energy for the K shell is 69.5 keV, so the energy of the K shell is –69.5 keV.



ANS. FIG. P42.54

The photon energies are

$$E = \frac{hc}{\lambda} = \frac{1\ 240\ \text{eV} \cdot \text{nm}}{\lambda} = \frac{1.240\ \text{keV} \cdot \text{nm}}{\lambda}$$

λ (nm)	photon energy (keV)	transition	energy of level (keV)	level
$\lambda_1 = 0.018\ 5$	67.03	$n = 4 \rightarrow 1$	–69.5 + 67.03 = –2.47	N
$\lambda_2 = 0.020\ 9$	59.3	$n = 3 \rightarrow 1$	–69.5 + 59.3 = –10.2	M
$\lambda_3 = 0.021\ 5$	57.7	$n = 2 \rightarrow 1$	–69.5 + 57.7 = –11.8	L

The ionization energy for the K shell is 69.5 keV, so the ionization energies for the other shells are:

$$\boxed{\text{L shell} = 11.8\ \text{keV}} \quad \boxed{\text{M shell} = 10.2\ \text{keV}} \quad \boxed{\text{N shell} = 2.47\ \text{keV}}$$

P42.55 Following the reasoning of Example 42.5, when the electron is in the K shell ($n = 1$), from Equation 42.37, its energy is

$$E_K \approx -Z^2 (13.6\ \text{eV})$$

When the electron was in the L shell ($n = 2$), the nuclear charge is shielded by one electron in the K shell, so (from page 1324)

$$E_L \approx -(Z-1)^2 \frac{13.6\ \text{eV}}{2^2}$$

When the electron drops from the L to the K shell of the atom, it emits a photon of energy (for $Z = 42$)

$$E_{\text{photon}} = E_L - E_K \approx (13.6\ \text{eV})\left[-\frac{(41)^2}{4} + (42)^2\right] = 1.83 \times 10^4\ \text{eV}$$

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with wavelength

$$\lambda = \frac{hc}{E_{\text{photon}}} = \frac{1\ 240\ \text{eV}\cdot\text{nm}}{1.83\times10^{4}\ \text{eV}} = 6.79\times10^{-2}\ \text{nm} = \boxed{0.068\ \text{nm}}$$

P42.56 (a) All of the kinetic energy of an electron after its acceleration through a potential difference ΔV goes into producing a single photon:

$$E = \frac{hc}{\lambda} = e\Delta V \quad \rightarrow \quad \Delta V = \frac{hc}{e\lambda} = \frac{1240\ \text{eV}\cdot\text{nm}}{\lambda} = \boxed{\frac{1240\ \text{V}\cdot\text{nm}}{\lambda}}$$

(b) The potential difference is inversely proportional to the wavelength.

(c) Yes. It predicts a minimum wavelength of 33.5 pm when the accelerating voltage is 37 keV, in agreement with the minimum wavelength in the figure.

(d) Yes, but it might be unlikely for a very high energy electron to stop in a single interaction to produce a high-energy gamma ray, and it might be difficult to observe the very low intensity radio waves produced as bremsstrahlung by low-energy electrons.

(e) The potential difference goes to infinity as the wavelength goes to zero.

(f) The potential difference goes to zero as the wavelength goes to infinity.

P42.57 The concepts for this problem are discussed in Example 42.5. An electron makes a transition from the M to the K shell. From Equation 42.37, when the electron is in the K shell, its energy is

$$E_{\text{K}} \approx -Z^{2}(13.6\ \text{eV})$$

When the electron was in the M shell, because nine electrons shield the nuclear charge—one in the L shell ($n = 1$) and eight in the K shell ($n = 2$)—its energy is

$$E_{\text{M}} \approx -(Z-9)^{2}\frac{13.6\ \text{eV}}{(3)^{2}}$$

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Thus, as the electron drops from the M to the K shell, it emits a photon of energy

$$E_{\text{photon}} = E_{\text{M}} - E_{\text{K}} \approx (13.6\text{ eV})\left[-\frac{(Z-9)^2}{9} + Z^2\right]$$

$$= (13.6\text{ eV})\left[-\left(\frac{Z^2 - 18Z + 81}{9}\right) + Z^2\right]$$

$$= (13.6\text{ eV})\left(\frac{8}{9}Z^2 + 2Z - 9\right) = \frac{hc}{\lambda}$$

Therefore, we have the relation

$$(13.6\text{ eV})\left(\frac{8}{9}Z^2 + 2Z - 9\right) = \frac{hc}{\lambda}$$

$$\frac{8}{9}Z^2 + 2Z - 9 = \frac{hc}{(13.6\text{ eV})\lambda} = \frac{(1\,240\text{ eV}\cdot\text{nm})}{(13.6\text{ eV})(0.101\text{ nm})} = 902.7$$

$$\frac{8}{9}Z^2 + 2Z - 911.7 = 0$$

Solving for Z gives

$$Z = \frac{-(2) \pm \sqrt{(2)^2 - 4(8/9)(-911.7)}}{2(8/9)} = \frac{-1 \pm \sqrt{1 + (8/9)(911.7)}}{(8/9)}$$

$$= \frac{-1 \pm 28.5}{(8/9)}$$

The positive solution is physical:

$$Z = \frac{-1 + 28.5}{(8/9)} = 30.9$$

The nearest whole number for Z is 31, which corresponds to the element $\boxed{\text{gallium}}$.

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Section 42.9 Spontaneous and Stimulated Transitions

Section 42.10 Lasers

P42.58 The electron in the E_3^* state drops to the E_2 state, emitting a photon of energy hf. The process conserves energy:

$$E_3^* = E_2 + hf \quad \rightarrow \qquad hf = E_3^* - E_2$$

The photon's energy is

$$hf = E_3^* - E_2 = (20.66 - 18.70)\ \text{eV} = 1.96\ \text{eV} = \frac{hc}{\lambda}$$

and its wavelength is

$$\lambda = \frac{hc}{E} = \frac{1\ 240\ \text{eV} \cdot \text{nm}}{1.96\ \text{eV}} = \boxed{633\ \text{nm}}$$

P42.59 (a) We use Equation 42.5, $E = hf = 0.117$ eV, and solve for f:

$$f = \frac{E}{h} = \frac{0.117\ \text{eV}}{6.626 \times 10^{-34}\ \text{J} \cdot \text{s}}\left(\frac{1.60 \times 10^{-19}\ \text{J}}{\text{eV}}\right)$$

$$= 2.83 \times 10^{13}\ \text{s}^{-1} = 28.3 \times 10^{12}\ \text{s}^{-1} = \boxed{28.3\ \text{THz}}$$

(b) The wavelength of the laser is

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^{8}\ \text{m/s}}{2.83 \times 10^{13}\ \text{s}^{-1}} = \boxed{10.6\ \mu\text{m}}$$

(c) This is in the $\boxed{\text{infrared}}$ portion of the electromagnetic spectrum.

P42.60 (a) We find the energy from

$$\Delta E = E_4^* - E_2 = \frac{hc}{\lambda}$$

Then,

$$E_2 = E_4^* - \frac{hc}{\lambda}$$

$$= 20.66\ \text{eV} - \frac{(6.626 \times 10^{-34}\ \text{J} \cdot \text{s})(3.00 \times 10^{8}\ \text{m/s})}{543 \times 10^{-9}\ \text{m}}\left(\frac{1\ \text{eV}}{1.602 \times 10^{-19}\ \text{J}}\right)$$

$$= \boxed{18.37\ \text{eV}}$$

(b) The light in the cavity is incident perpendicularly on the mirrors. Some of the light reflects off the front surface of the silicon dioxide layer while some enters the layer and then reflects off the titanium oxide layer on its other side. The index of refraction of

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titanium oxide (1.9–2.6) is greater the index of refraction of silicon dioxide (1.458), so there is automatically a 180° shift of the ray reflecting off the silicon dioxide. To minimize reflection at a vacuum wavelength of 632.8 nm, the net phase difference between reflected rays should be 180°, so the extra distance traveled by the ray passing into the silicon dioxide should be one whole wavelength:

$$2t = \frac{\lambda}{n}$$

$$t = \frac{\lambda}{2n} = \frac{632.8 \text{ nm}}{2(1.458)} = \boxed{217 \text{ nm}}$$

(c) For the green light to experience constructive interference, the net phase difference should be 360°, including contributions of 180° by reflection and 180° by extra distance traveled:

$$2t = \frac{\lambda}{2n}$$

$$t = \frac{\lambda}{4n} = \frac{543 \text{ nm}}{4(1.458)} = \boxed{93.1 \text{ nm}}$$

P42.61 The energy in each pulse is

$$E = P\Delta t = \left(1.00 \times 10^{6} \text{ W}\right)\left(1.00 \times 10^{-8} \text{ s}\right) = 1.00 \times 10^{-2} \text{ J}$$

The energy of each photon is

$$E_\gamma = hf = \frac{hc}{\lambda} = \frac{\left(6.626 \times 10^{-34} \text{ J} \cdot \text{s}\right)\left(3.00 \times 10^{8} \text{ m/s}\right)}{694.3 \times 10^{-9} \text{ m}}$$

$$= 2.86 \times 10^{-19} \text{ J}$$

So the number of photons in the pulse is

$$N = \frac{E}{E_\gamma} = \frac{1.00 \times 10^{-2} \text{ J}}{2.86 \times 10^{-19} \text{ J/photon}} = \boxed{3.49 \times 10^{16} \text{ photons}}$$

P42.62 (a) The equilibrium ratio is

$$\frac{N_4^*}{N_3} = \frac{N_g e^{-E_3/k_B T}}{N_g e^{-E_2/k_B T}} = e^{-(E_3 - E_2)/k_B T} = e^{-\Delta E/k_B T}$$

where the temperature T = 27.0 °C + 273.15 = 300.2 K, and the energy difference (from Figure P42.60) is

$$\Delta E = E_4^* - E_3 = 20.66 \text{ eV} - 18.70 \text{ eV} = 1.96 \text{ eV}$$

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Substituting numerical values,

$$\frac{N_4^*}{N_3} = e^{-\Delta E/k_B T} = e^{-(1.96\text{ eV})(1.602\times10^{-19}\text{ J/eV})/(1.381\times10^{-23}\text{ J/K})(300.2\text{ K})}$$

$$= \boxed{1.26\times10^{-33}}$$

(b) Now, we require $\frac{N_4^*}{N_3} = e^{-\Delta E/k_B T} = 1.02$

where $\Delta E = E_4^* - E_3 = 1.96\text{ eV}$

Thus,

$$\ln(1.02) = -\frac{(1.96\text{ eV})(1.602\times10^{-19}\text{ J/eV})}{(1.381\times10^{-23}\text{ J/K})T}$$

$$T = \boxed{-1.15\times10^{6}\text{ K}}$$

(c) The population inversion requires the temperature be negative. Because $\Delta E = E_4^* - E_3 > 0$, and in any real equilibrium state $T > 0$, the ratio $N_4^*/N_3 = e^{-\Delta E/k_B T} < 1$. Thus, a population inversion cannot happen in thermal equilibrium.

P42.63 (a) The energy of the pulse is spread over the area of a circle of radius $R = 15.0\ \mu\text{m}$:

$$I = \frac{P}{A} = \frac{\Delta E/\Delta t}{\pi R^2} = \frac{3.00\times10^{-3}\text{ J}/1.00\times10^{-9}\text{ s}}{\left[\pi(15.0\times10^{-6}\text{ m})^2\right]}$$

$$= \boxed{4.24\times10^{15}\text{ W/m}^2}$$

(b) The absorbed energy falls within the area of a circle of radius $r = 0.300$ nm:

$$\Delta E = I\Delta t A = \left(\frac{\Delta E/\Delta t}{\pi R^2}\right)\Delta t \pi r^2 = \Delta E\left(\frac{r^2}{R^2}\right)$$

$$= (3.00\times10^{-3}\text{ J})\left[\frac{(0.300\times10^{-9}\text{ m})^2}{(15.0\times10^{-6}\text{ m})^2}\right] = \boxed{1.20\times10^{-12}\text{ J}}$$

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***P42.64** (a) The distance between nodes is $\frac{\lambda}{2}$, so we require solutions to $35.124\ 103\ \text{cm} = \frac{N}{2}\lambda$, where N is an integer and λ is in the required range. The midpoint of the range is 632.809 10 nm, giving

$$N_{\text{trial}} = \frac{2(35.124\ 103 \times 10^{-2}\ \text{m})}{632.809\ 1 \times 10^{9}\ \text{m}} = 1\ 110\ 101.07$$

So we try N = 1 110 101, 1 110 102, 1 110 100, 1 110 103, and so on:

$$\lambda_1 = \frac{2(35.124\ 103 \times 10^{-2}\ \text{m})}{1\ 110\ 101} = \boxed{632.809\ 14\ \text{nm}}$$

$$\lambda_2 = \frac{2(35.124\ 103 \times 10^{-2}\ \text{m})}{1\ 110\ 102} = \boxed{632.808\ 57\ \text{nm}}$$

$$\lambda_3 = \frac{2(35.124\ 103 \times 10^{-2}\ \text{m})}{1\ 110\ 100} = \boxed{632.809\ 71\ \text{nm}}$$

$$\lambda_{\text{trial}} = \frac{2(35.124\ 103 \times 10^{-2}\ \text{m})}{1\ 110\ 103} = 632.808\ 00\ \text{nm}$$

outside the range. Thus the laser light has just $\boxed{\text{three}}$ wavelength components.

(b) The rms speed is obtained from $\frac{1}{2}m_0 v^2 = \frac{3}{2}kT$. We use the periodic table for the mass of a neon atom. Then,

$$v = \sqrt{\frac{3kT}{m_0}} = \sqrt{\frac{3(1.38 \times 10^{-23}\ \text{J/K})(393\ \text{K})}{20.18\ \text{u}}\left(\frac{1\ \text{u}}{1.66 \times 10^{-27}\ \text{kg}}\right)}$$

$$= \boxed{697\ \text{m/s}}$$

(c) For a neon atom moving toward one mirror at the rms speed as it emits, the Doppler shift is described by

$$f' = f\sqrt{\frac{c+v}{c-v}} = \frac{c}{\lambda'} = \frac{c}{\lambda}\sqrt{\frac{c+v}{c-v}}$$

$$\lambda' = \lambda\sqrt{\frac{c-v}{c+v}} = (632.809\ 1\ \text{nm})\sqrt{\frac{3 \times 10^8 - 697}{3 \times 10^8 + 697}} = 632.807\ 63\ \text{nm}$$

This is outside the given range. Many atoms are moving faster than the rms speed, so we should expect still more Doppler broadening of the resonance amplification peak.

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Additional Problems

P42.65 To ionize the atom, it is necessary that $n_f \to \infty$. The required energy is then

$$\Delta E = E_f - E_i = -13.6\text{ eV}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) = -13.6\text{ eV}\left(\frac{1}{\infty} - \frac{1}{n_i^2}\right)$$

$$= \frac{13.6\text{ eV}}{n_i^2}$$

(a) If $n_i = 1$, the required energy is

$$\Delta E = \frac{13.6\text{ eV}}{1^2} = \boxed{13.6\text{ eV}}$$

(b) If $n_i = 3$,

$$\Delta E = \frac{13.6\text{ eV}}{3^2} = \boxed{1.51\text{ eV}}$$

P42.66 The sliver atoms ($Z = 108$) move as particles initially traveling in the x direction at speed u with constant acceleration in the z direction:

$$\Delta z = \frac{1}{2}a(\Delta t)^2 = \frac{1}{2}\left(\frac{F_z}{m_0}\right)(\Delta t)^2 = \frac{\mu_z\left(dB_z/dz\right)}{2m_0}\left(\frac{\Delta x}{u}\right)^2$$

where $\mu_z = \dfrac{e\hbar}{2m_e}$, so

$$\Delta z = \frac{e\hbar}{4m_e m_0}\left(\frac{\Delta x}{u}\right)^2 \frac{dB_z}{dz}$$

Therefore,

$$\frac{dB_z}{dz} = \frac{4m_0 m_e \Delta z v^2}{e\hbar \Delta x^2}$$

$$= \frac{4\left[(108)\left(1.66\times10^{-27}\text{ kg}\right)\right]\left(9.11\times10^{-31}\text{ kg}\right)\left(10^{-3}\text{ m}\right)(100\text{ m/s})^2}{\left(1.60\times10^{-19}\text{ C}\right)\left(1.05\times10^{-34}\text{ J}\cdot\text{s}\right)(1.00\text{ m})^2}$$

$$\frac{dB_z}{dz} = \boxed{0.389\text{ T/m}}$$

P42.67 The wave function for the 2*s* state is given by Equation 42.26:

$$\psi_{2s}(r) = \frac{1}{4\sqrt{2\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left[2 - \frac{r}{a_0}\right]e^{-r/2a_0}$$

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(a) Taking $r = a_0 = 0.529 \times 10^{-10}\ \text{m},$ we find

$$\psi_{2s}(a_0) = \frac{1}{4\sqrt{2\pi}}\left(\frac{1}{0.529 \times 10^{-10}\ \text{m}}\right)^{3/2}[2-1]e^{-1/2}$$

$$= \boxed{1.57 \times 10^{14}\ \text{m}^{-3/2}}$$

(b) $\left|\psi_{2s}(a_0)\right|^2 = \left(1.57 \times 10^{14}\ \text{m}^{-3/2}\right)^2 = \boxed{2.47 \times 10^{28}\ \text{m}^{-3}}$

(c) Using Equation 42.24 and the result of part (b) gives

$$P_{2s}(a_0) = 4\pi a_0^2 \left|\psi_{2s}(a_0)\right|^2 = \boxed{8.69 \times 10^{8}\ \text{m}^{-1}}$$

P42.68 (a) The energy difference between these two states is equal to the energy that is absorbed. Thus,

$$\Delta E = E_2 - E_1 = \frac{(-13.6\ \text{eV})}{4} - \frac{(-13.6\ \text{eV})}{1} = 10.2\ \text{eV}$$

$$= \boxed{1.63 \times 10^{-18}\ \text{J}}$$

(b) $E = \frac{3}{2}k_B T \quad \text{or} \quad T = \frac{2E}{3k_B} = \frac{2\left(1.63 \times 10^{-18}\ \text{J}\right)}{3\left(1.38 \times 10^{-23}\ \text{J/K}\right)} = \boxed{7.88 \times 10^{4}\ \text{K}}$

P42.69 To evaluate the energy difference, we imagine the z component of the electron's magnetic moment as continuously variable. In turning it from alignment with the field to the opposite direction, the field does work according to Equation 10.22,

$$W = \int dW = \int_0^{180^\circ} \tau\ d\theta = \int_0^{\pi} \mu B \sin\theta\ d\theta = -\mu B \cos\theta \Big|_0^{\pi} = 2\mu B$$

To make the electron flip, the photon must carry energy $\Delta E = 2\mu_B B = hf$. Therefore,

$$f = \frac{2\mu_B B}{h} = \frac{2\left(9.27 \times 10^{-24}\ \text{J/T}\right)(0.350\ \text{T})}{6.626 \times 10^{-34}\ \text{J}\cdot\text{s}}$$

$$= 9.79 \times 10^{9}\ \text{Hz} = \boxed{9.79\ \text{GHz}}$$

P42.70 We suppose that the electron that makes the transition is shielded from the electric field of the full nuclear charge by the one K-shell electron originally below it. With $Z = 24$, its original energy is

$$E = -(Z-1)^2(13.6\ \text{eV})\left(\frac{1}{2^2}\right) = -1.80\ \text{keV}$$

Its final energy is $E = -Z^2(13.6\ \text{eV})\left(\frac{1}{1^2}\right) = -7.83\ \text{keV}.$

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The magnitude of the electron's energy loss is

$$7.83\text{ keV} - 1.80\text{ keV} = 6.04\text{ keV}$$

Then, instead of coming out as an x-ray photon, this +6.04 keV can be transferred to the single 4*s* electron. Suppose that it is shielded by the 22 electrons in the K, L, and M shells. To break the outermost electron out of the atom, producing a Cr^{2+} ion, requires an energy investment of

$$E_{\text{ionize}} = \frac{(Z-22)^2(13.6\text{ eV})}{4^2} = \frac{2^2(13.6\text{ eV})}{16} = 3.40\text{ eV}$$

Then the remaining energy that can appear as kinetic energy is

$$K = |\Delta E| - E_{\text{ionize}} = 6\ 035\text{ eV} - 3.4\text{ eV} = \boxed{6.03\text{ keV}}$$

Because of conservation of momentum for the ion-electron system and the tiny mass of the electron compared to that of the Cr^{2+} ion, almost all of this kinetic energy will belong to the electron.

***P42.71** From Figure 42.20, a typical ionization energy is 8 eV. For internal energy to ionize most of the atoms we require

$$\frac{3}{2}k_BT = 8\text{ eV}:$$

$$T = \frac{2\times 8\left(1.60\times 10^{-19}\text{ J}\right)}{3\left(1.38\times 10^{-23}\text{ J/K}\right)} \boxed{\sim \text{ between } 10^4\text{ K and } 10^5\text{ K}}$$

***P40.72** From Equation 42.26,

$$\psi_{2s} = \frac{1}{4}(2\pi)^{-1/2}\left(\frac{1}{a_0}\right)^{3/2}\left(2-\frac{r}{a_0}\right)e^{-r/2a_0} = A\left(2-\frac{r}{a_0}\right)e^{-r/2a_0}$$

Differentiating gives

$$\frac{d\psi}{dr} = Ae^{-r/2a_0}\left(-\frac{2}{a_0}+\frac{r}{2a_0^2}\right)$$

Differentiating a second time gives,

$$\frac{d^2\psi}{dr^2} = \left(\frac{Ae^{-r/2a_0}}{a_0^2}\right)\left(\frac{3}{2}-\frac{r}{4a_0}\right)$$

Substituting into Schrödinger's equation and dividing by $Ae^{-r/2a_0}$, we will have a solution if

$$-\frac{5}{4}\frac{\hbar^2}{m_e a_0^2}+\frac{k_e e^2}{a_0}+\frac{\hbar^2 r}{8m_e a_0^3}+\frac{2\hbar^2}{m_e a_0 r}-\frac{2k_e e^2}{r} = 2E-\frac{Er}{a_0}$$

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Now with $a_0 = \dfrac{\hbar^2}{m_e e^2 k_e}$, this reduces to

$$-\frac{m_e e^4 k_e^2}{8\hbar^2}\left(2 - \frac{r}{a_0}\right) = E\left(2 - \frac{r}{a_0}\right)$$

This is true, so ψ_{2s} is a solution to the Schrödinger equation, provided $E = \dfrac{1}{4}E_1 = -3.40$ eV.

***P42.73** The expectation value of $1/r$ is found from

$$\left\langle \frac{1}{r} \right\rangle = \int_0^\infty \frac{4r^2}{a_0^3} e^{-2r/a_0} \frac{1}{r} dr = \frac{4}{a_0^3}\int_0^\infty r e^{-(2/a_0)r}\, dr = \frac{4}{a_0^3}\frac{1}{\left(\frac{2}{a_0}\right)^2} = \boxed{\frac{1}{a_0}}$$

We compare this to $\dfrac{1}{\langle r \rangle} = \dfrac{1}{\frac{3a_0}{2}} = \dfrac{2}{3a_0}$, and find that the average reciprocal value is $\boxed{\text{NOT}}$ the reciprocal of the average value.

P42.74 The fact that there are five values of the z component of orbital angular momentum tells us that there are five values of m_ℓ, which, in turn, tells us that $\ell = 2$. From Equation 42.28, we can find the maximum value of m_ℓ:

$$L_z = m_\ell \hbar \qquad \rightarrow \qquad m_\ell = \frac{L_z}{\hbar} = \frac{3.16 \times 10^{-34}}{1.055 \times 10^{-34}} = 3$$

In order to have a maximum value of m_ℓ equal to 3, we need to have $\ell = 3$, which is inconsistent with the first result.

P42.75 (a) The size of the quantum jump in the electron's energy is

$$\Delta E = \frac{e\hbar B}{m_e} = \frac{(1.60\times10^{-19}\ \text{C})(6.626\times10^{-34}\ \text{J}\cdot\text{s})(5.26\ \text{T})}{2\pi(9.11\times10^{-31}\ \text{kg})}$$

$$= 9.75\times10^{-23}\ \text{J}\left(\frac{1\ \text{eV}}{1.60\times10^{-19}\ \text{J}}\right) = 6.09\times10^{-4}\ \text{eV} = \boxed{609\ \mu\text{eV}}$$

(b) The energy available from the walls of the container is

$$k_B T = (1.38\times10^{-23}\ \text{J/K})(80\times10^{-3}\ \text{K}) = 1.10\times10^{-24}\ \text{J} = \boxed{6.9\ \mu\text{eV}}$$

(c) The photon's frequency is

$$f = \frac{\Delta E}{h} = \frac{9.75\times10^{-23}\ \text{J}}{6.626\times10^{-34}\ \text{J}\cdot\text{s}} = 1.47\times10^{11}\ \text{Hz} = 147\times10^{9}\ \text{Hz}$$

$$= \boxed{147\ \text{GHz}}$$

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(d) The photon's wavelength is

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8 \text{ m/s}}{1.47 \times 10^{11} \text{ Hz}} = 2.04 \times 10^{-3} \text{ m} = \boxed{2.04 \text{ mm}}$$

P42.76 (a) Using the same procedure that was used in the Bohr model of the hydrogen atom, we apply Newton's second law to the Earth. We simply replace the Coulomb force by the gravitational force exerted by the Sun on the Earth and find

$$G\frac{M_S M_E}{r^2} = M_E \frac{v^2}{r} \qquad \textbf{[1]}$$

where v is the orbital speed of the Earth. Next, we apply the postulate that angular momentum of the Earth is quantized in multiples of $\hbar$:

$$M_E v r = n\hbar \qquad (n = 1,\ 2,\ 3,\ \ldots)$$

Solving for v gives

$$v = \frac{n\hbar}{M_E r} \qquad \textbf{[2]}$$

Substituting [2] into [1], we find

$$r = \frac{n^2\hbar^2}{GM_S M_E^2} \qquad \textbf{[3]}$$

(b) Solving equation [3] for n gives

$$n = \sqrt{GM_S r}\,\frac{M_E}{\hbar} \qquad \textbf{[4]}$$

Taking $M_S = 1.99 \times 10^{30}$ kg, $M_E = 5.98 \times 10^{24}$ kg, $r = 1.496 \times 10^{11}$ m, $G = 6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2/\text{kg}^2$, and $\hbar = 1.055 \times 10^{-34}\ \text{J} \cdot \text{s}$, we find

$$n = \boxed{2.53 \times 10^{74}}$$

(c) We can use equation [3] to determine the radii for the orbits corresponding to the quantum numbers n and $n + 1$:

$$r_n = \frac{n^2\hbar^2}{GM_S M_E^2} \quad \text{and} \quad r_{n+1} = \frac{(n+1)^2\hbar^2}{GM_S M_E^2}$$

Hence, the separation between these two orbits is

$$\Delta r = \frac{\hbar^2}{GM_S M_E^2}\left[(n+1)^2 - n^2\right] = \frac{\hbar^2}{GM_S M_E^2}(2n+1)$$

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Since n is very large, we can neglect the number 1 in the parentheses and express the separation as

$$\Delta r \approx \frac{\hbar^2}{GM_S M_E^2}(2n)$$

$$= \frac{\left(1.054\ 6\times10^{-34}\ \text{J}\cdot\text{s}\right)^2}{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(1.99\times10^{30}\ \text{kg}\right)\left(5.98\times10^{24}\ \text{kg}\right)^2} \times\left[2\left(2.53\times10^{74}\right)\right]$$

$$= \boxed{1.18\times10^{-63}\ \text{m}}$$

(d) This number is *much smaller* than the radius of an atomic nucleus ($\sim 10^{-15}$ m), so the distance between quantized orbits of the Earth is too small to observe.

P42.77 The average squared separation distance is

$$\left\langle r^2\right\rangle = \int_{\text{all space}} \psi_{1s}^* r^2 \psi_{1s}\, dV$$

$$= \int_{r=0}^{\infty}\left(\frac{1}{\sqrt{\pi a_0^3}}e^{-r/a_0}\right) r^2 \left(\frac{1}{\sqrt{\pi a_0^3}}e^{-r/a_0}\right) 4\pi r^2\, dr$$

$$= \frac{4}{a_0^3}\int_0^{\infty} r^4 e^{-2r/a_0}\, dr$$

We use $\int_0^{\infty} x^n e^{-ax} dx = \dfrac{n!}{a^{n+1}}$ from Table B.6:

$$\left\langle r^2\right\rangle = \frac{4}{a_0^3}\frac{4!}{\left(2/a_0\right)^5} = \frac{96a_0^2}{32} = 3a_0^2$$

The root-mean-square uncertainty in r is

$$\Delta r = \sqrt{\left\langle r^2\right\rangle - \left\langle r\right\rangle^2} = \left[3a_0^2 - \left(\frac{3a_0}{2}\right)^2\right]^{1/2}$$

$$= \left(3a_0^2 - \frac{9a_0^2}{4}\right)^{1/2} = \left(\frac{3}{4}\right)^{1/2} a_0 = \boxed{0.866a_0}$$

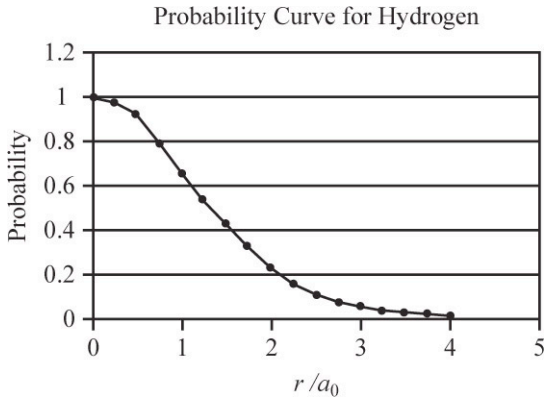
P42.78 (a) From Equations 42.22 – 42.25,

$$P = \int_r^{\infty} P_{1s}(r')\, dr' = \frac{4}{a_0^3}\int_r^{\infty} r'^2 e^{-2r'/a_0}\, dr'$$

$$= \left[-\left(\frac{2r'^2}{a_0^2} + \frac{2r'}{a_0} + 1\right)e^{-2r'/a_0}\right]_r^{\infty} = \boxed{\left(\frac{2r^2}{a_0^2} + \frac{2r}{a_0} + 1\right)e^{-2r/a_0}}$$

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(b) The graph is shown in ANS. FIG. P42.78.



ANS. FIG. P42.78

(c) The probability of finding the electron inside or outside the sphere of radius r is $\frac{1}{2}$:

$$\left(\frac{2r^2}{a_0^2}+\frac{2r}{a_0}+1\right)e^{-2r/a_0}=\frac{1}{2} \quad \text{or} \quad z^2+2z+2=e^z$$

where $z=\frac{2r}{a_0}$.

One can home in on a solution to this transcendental equation for r on a calculator, the result being $r=\boxed{1.34a_0}$ to three digits.

P42.79 (a) The energy emitted by the atom is

$$\Delta E=E_4-E_2=-13.6\text{ eV}\left(\frac{1}{4^2}-\frac{1}{2^2}\right)=2.55\text{ eV}$$

The wavelength of the photon produced is then

$$\lambda=\frac{hc}{E_\gamma}=\frac{hc}{\Delta E}=\frac{1\,240\text{ eV}\cdot\text{nm}}{2.55\text{ nm}}=\boxed{486\text{ nm}}$$

(b) Since momentum must be conserved, the photon and the atom go in opposite directions with equal magnitude momenta. Thus, $p=m_{\text{atom}}v=\frac{h}{\lambda}$, or

$$v=\frac{h}{m_{\text{atom}}\lambda}=\frac{6.626\times10^{-34}\text{ J}\cdot\text{s}}{\left(1.67\times10^{-27}\text{ kg}\right)\left(4.86\times10^{-7}\text{ m}\right)}$$

$$=\boxed{0.816\text{ m/s}}$$

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P42.80 (a) The energy of the ground state is

$$E_1 = -\frac{hc}{\lambda_{\text{series limit}}} = -\frac{1\ 240\ \text{eV}\cdot\text{nm}}{152.0\ \text{nm}} = \boxed{-8.16\ \text{eV}}$$

From the wavelength of the Lyman α line:

$$E_2 - E_1 = \frac{hc}{\lambda} = \frac{1\ 240\ \text{nm}\cdot\text{eV}}{202.6\ \text{nm}} = 6.12\ \text{eV}$$

$$E_2 = E_1 + 6.12\ \text{eV} = \boxed{-2.04\ \text{eV}}$$

The wavelength of the Lyman β line gives

$$E_3 - E_1 = \frac{1\ 240\ \text{nm}\cdot\text{eV}}{170.9\ \text{nm}} = 7.26\ \text{eV}$$

so $E_3 = \boxed{-0.902\ \text{eV}}$.

Next, using the Lyman γ line gives

$$E_4 - E_1 = \frac{1\ 240\ \text{nm}\cdot\text{eV}}{162.1\ \text{nm}} = 7.65\ \text{eV}$$

and $E_4 = \boxed{-0.508\ \text{eV}}$.

From the Lyman δ line,

$$E_5 - E_1 = \frac{1\ 240\ \text{nm}\cdot\text{eV}}{158.3\ \text{nm}} = 7.83\ \text{eV}$$

so $E_5 = \boxed{-0.325\ \text{eV}}$.

(b) For the Balmer series,

$$\frac{hc}{\lambda} = E_i - E_2, \text{ or } \lambda = \frac{1\ 240\ \text{nm}\cdot\text{eV}}{E_i - E_2}$$

For the α line, $E_i = E_3$ and so

$$\lambda_a = \frac{1\ 240\ \text{nm}\cdot\text{eV}}{(-0.902\ \text{eV}) - (-2.04\ \text{eV})} = \boxed{1\ 090\ \text{nm}}$$

Similarly, the wavelengths of the β line, γ line, and the short wavelength limit are found to be: $\boxed{811\ \text{nm, } 724\ \text{nm, and } 609\ \text{nm}}$.

(c) Using Equation 42.2, $\frac{1}{\lambda} = R_{\text{H}}\left(1 - \frac{1}{n^2}\right)$, the Lyman series for hydrogen contains the lines: $\alpha\,(n = 2) = 122$ nm, $\beta\,(n = 3) = 103$ nm, $\gamma(n = 4) = 97.2$ nm, $\delta\,(n = 5) = 94.9$ nm, the short wavelength limit ($n \to \infty$) = 91.1 nm.

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Computing 60.0% of the wavelengths of the spectral lines shown on the energy-level diagram gives:

$$0.600(202.6\text{ nm}) = \boxed{122\text{ nm}}$$

$$0.600(170.9\text{ nm}) = \boxed{103\text{ nm}}$$

$$0.600(162.1\text{ nm}) = \boxed{97.3\text{ nm}}$$

$$0.600(158.3\text{ nm}) = \boxed{95.0\text{ nm}}$$

$$0.600(152.0\text{ nm}) = \boxed{91.2\text{ nm}}$$

These are seen to be the wavelengths of the α, β, γ, and δ lines as well as the short wavelength limit for the Lyman series in Hydrogen.

(d) The observed wavelengths could be the result of Doppler shift when the source moves away from the Earth. The required speed of the source is found from

$$\frac{f'}{f} = \frac{\lambda}{\lambda'} = \sqrt{\frac{c-v}{c+v}} = 0.600 \qquad \text{yielding} \quad v = 0.471c.$$

The spectrum could be that of hydrogen, Doppler-shifted by motion away from us at speed $0.471c$.

P42.81 We use Equation 42.26:

$$\psi_{2s}(r) = \frac{1}{4\sqrt{2\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left[2 - \frac{r}{a_0}\right]e^{-r/2a_0}$$

(a) By Equation 42.24,

$$P(r) = 4\pi r^2|\psi|^2 = \boxed{\frac{r^2}{8a_0^3}\left(2-\frac{r}{a_0}\right)^2 e^{-r/a_0}}$$

(b) The derivative of the radial probability is

$$\frac{dP(r)}{dr} = \frac{1}{8a_0^3}\left[2r\left(2-\frac{r}{a_0}\right)^2 - 2r^2\left(\frac{1}{a_0}\right)\left(2-\frac{r}{a_0}\right) - r^2\left(2-\frac{r}{a_0}\right)^2\left(\frac{1}{a_0}\right)\right]e^{-r/a_0}$$

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Simplifying the expression,

$$\frac{dP(r)}{dr} = \frac{1}{8a_0^3}\left(2-\frac{r}{a_0}\right)e^{-r/a_0}\left[4r - \frac{2r^2}{a_0} - \frac{2r^2}{a_0} - \left(\frac{2r^2}{a_0} - \frac{r^3}{a_0^2}\right)\right]$$

$$= \frac{r}{8a_0^5}\left(2-\frac{r}{a_0}\right)e^{-r/a_0}\left[4a_0^2 - 6ra_0 + r^2\right]$$

$$= \boxed{\frac{r}{8a_0^5}\left(2-\frac{r}{a_0}\right)e^{-r/a_0}\left[r^2 - 6ra_0 + 4a_0^2\right]}$$

(c) Its extremes are given by

$$\frac{dP}{dr} = \frac{r}{8a_0^5}\left(2-\frac{r}{a_0}\right)e^{-r/a_0}\left[r^2 - 6ra_0 + 4a_0^2\right] = 0$$

The roots of $\frac{dP}{dr} = 0$ at $\boxed{r = 0,\ r = 2a_0, \text{ and } r = \infty}$ are minima with $P(r) = 0$ (as shown in Figure 42.12).

(d) We require $r^2 - 6ra_0 + 4a_0^2 = 0$. The solutions are

$$r = \frac{-(-6a_0) \pm \sqrt{(-6a_0)^2 - 4(1)(4a_0^2)}}{2} = \frac{6a_0 \pm \sqrt{20a_0^2}}{2} = \boxed{\left(3 \pm \sqrt{5}\right)a_0}$$

(e) We substitute the last two roots into $P(r)$ to determine the most probable value:

When $r = \left(3-\sqrt{5}\right)a_0 = 0.764\,a_0$,

$$P(r) = \frac{(0.764\,a_0)^2}{8a_0^3}\left(2 - \frac{0.764\,a_0}{a_0}\right)^2 e^{-0.764}$$

$$= \frac{(0.764)^2}{8a_0}(2-0.764)^2\,e^{-0.764} = \frac{0.051\,9}{a_0}$$

When $r = \left(3+\sqrt{5}\right)a_0 = 5.236a_0$,

$$P(r) = \frac{(5.236\,a_0)^2}{8a_0^3}\left(2 - \frac{5.236\,a_0}{a_0}\right)^2 e^{-5.236}$$

$$= \frac{(5.236)^2}{8a_0}(2-5.236)^2\,e^{-5.236} = \frac{0.191}{a_0}$$

Therefore, the most probable value of r is

$\boxed{r = \left(3+\sqrt{5}\right)a_0 \rightarrow P = 0.191/a_0}$.

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P42.82 (a) One molecule's share of volume is,

$$\text{Al: } V = \left(\frac{27.0\text{ g}}{\text{mol}}\right)\left(\frac{1\text{ mol}}{6.02\times10^{23}\text{ molecules}}\right)\left(\frac{1.00\times10^{-6}\text{ m}^3}{2.70\text{ g}}\right)$$

$$= 1.66\times10^{-29}\text{ m}^3$$

$$D \approx \sqrt[3]{V} = \boxed{2.55\times10^{-10}\text{ m}\sim10^{-1}\text{ nm}}$$

$$\text{U: } V = \left(\frac{238\text{ g}}{\text{mol}}\right)\left(\frac{1\text{ mol}}{6.02\times10^{23}\text{ molecules}}\right)\left(\frac{1.00\times10^{-6}\text{ m}^3}{18.9\text{ g}}\right)$$

$$= 2.09\times10^{-29}\text{ m}^3$$

$$D \approx \sqrt[3]{V} = \boxed{2.76\times10^{-10}\text{ m} \sim 10^{-1}\text{ nm}}$$

(b) The outermost electron in any atom sees the nuclear charge screened by all the electrons below it. If we can visualize a single outermost electron, it moves in the electric field of net charge $+Ze-(Z-1)e=+e$, the charge of a single proton, as felt by the electron in hydrogen. So the Bohr radius sets the scale for the outside diameter of every atom. An innermost electron, on the other hand, sees the nuclear charge unscreened, and the scale size of its (K-shell) orbit is a_0/Z.

P42.83 (a) The length of the pulse is

$$\Delta L = c\Delta t = \left(3.00\times10^{8}\text{ m/s}\right)\left(14.0\times10^{-12}\text{ s}\right) = \boxed{4.20\text{ mm}}$$

(b) The energy of each photon is

$$E = \frac{hc}{\lambda} = 2.86\times10^{-19}\text{ J}$$

so the number of photons in the pulse is

$$N = \frac{3.00\text{ J}}{2.86\times10^{-19}\text{ J/photon}} = \boxed{1.05\times10^{19}\text{ photons}}$$

(c) The volume of the pulse is

$$V = \Delta L\pi r^2 = (4.20\text{ mm})\left[\pi(3.00\text{ mm})^2\right] = 119\text{ mm}^3$$

resulting in a photon density of

$$n = \frac{1.05\times10^{19}\text{ photons}}{119\text{ mm}^3} = \boxed{8.82\times10^{16}\text{ mm}^{-3}}$$

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P42.84 (a) The length of the pulse is $\Delta L = \boxed{c\Delta t}$.

(b) The energy of each photon is $E = \dfrac{hc}{\lambda}$, so

$$N = \frac{T_{\text{ER}}}{E} = \boxed{\frac{\lambda T_{\text{ER}}}{hc}}$$

(c) The volume of the pulse is

$$V = \Delta L \pi \frac{d^2}{4} = c\Delta t \pi \frac{d^2}{4}$$

resulting in a photon density of

$$n = \frac{N}{V} = \frac{\lambda T_{\text{ER}}}{hc\left(c\Delta t \pi d^2/4\right)} = \boxed{\frac{4\lambda T_{\text{ER}}}{\pi h c^2 d^2 \Delta t}}$$

P42.85 The fermions are described by the exclusion principle. Two of them, one spin-up and one spin-down, will be in the ground energy level, in a standing wave pattern with one antinode:

$$d_{\text{NN}} = \frac{1}{2}\lambda = L \rightarrow \lambda = 2L = \frac{h}{p},$$

and $$p = \frac{h}{2L} \rightarrow K = \frac{1}{2}mv^2 = \frac{p^2}{2m} = \frac{h^2}{8mL^2}.$$

The third must be in the next higher level, in a standing wave pattern with two antinodes:

$$2d_{\text{NN}} = 2\left(\frac{\lambda}{2}\right) = L \rightarrow \lambda = L,$$

and $$p = \frac{h}{L} \rightarrow K = \frac{p^2}{2m} = \frac{h^2}{2mL^2}.$$

The total energy is then

$$\frac{h^2}{8mL^2} + \frac{h^2}{8mL^2} + \frac{h^2}{2mL^2} = \boxed{\frac{3h^2}{4mL^2}}$$

P42.86 An ionization energy of 4.10 eV means the ground state energy is –4.10 eV. The photon energies tell us the separation of the energy levels:

$$E = \frac{hc}{\lambda} = \frac{1\,240\ \text{eV}\cdot\text{nm}}{\lambda} = \Delta E$$

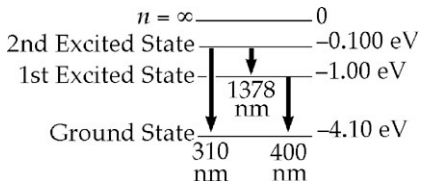
Then, $\lambda_1 = 310\ \text{nm}$, so $\Delta E_1 = 4.00\ \text{eV}$

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$$\lambda_2 = 400 \text{ nm}, \qquad \Delta E_2 = 3.10 \text{ eV}$$

$$\lambda_3 = 1\,378 \text{ nm}, \qquad \Delta E_3 = 0.900 \text{ eV}$$

The energy level diagram having the fewest levels and consistent with these energies is shown in ANS. FIG. P42.86.



ANS. FIG. 42.86

P42.87 The general radial probability distribution function is

$$P(r) = 4\pi r^2 |\psi|^2$$

With $\psi_{1s} = \left(\pi a_0^3\right)^{-1/2} e^{-r/a_0}$, it is $P(r) = 4r^2 a_0^{-3} e^{-2r/a_0}$.

The required probability is then

$$P = \int_{2.50a_0}^{\infty} P(r)\,dr = \int_{2.50a_0}^{\infty} \frac{4r^2}{a_0^3} e^{-2r/a_0}\,dr$$

Let $z = 2r/a_0$ and $dz = 2dr/a_0$. Then we want $P = \frac{1}{2}\int_{5.00}^{\infty} z^2 e^{-z}\,dz$.

Performing this integration by parts,

$$P = -\frac{1}{2}\left(z^2 + 2z + 2\right)e^{-z}\Big|_{5.00}^{\infty}$$

$$P = -\frac{1}{2}(0) + \frac{1}{2}(25.0 + 10.0 + 2.00)e^{-5.00} = \left(\frac{37}{2}\right)(0.006\,74) = \boxed{0.125}$$

P42.88 From Equations 42.22 – 42.25,

$$P = \int_{\beta a_0}^{\infty} P_{1s}(r)\,dr = \int_{\beta a_0}^{\infty} \frac{4r^2}{a_0^3} e^{-2r/a_0}\,dr = \frac{1}{2}\int_{2\beta}^{\infty} z^2 e^{-z}\,dz, \quad \text{where} \quad z \equiv \frac{2r}{a_0}$$

$$P = -\frac{1}{2}\left(z^2 + 2z + 2\right)e^{-z}\Big|_{2\beta}^{\infty} = -\frac{1}{2}[0] + \frac{1}{2}\left[(2\beta)^2 + 4\beta + 2\right]e^{-2\beta}$$

$$= \boxed{e^{-2\beta}\left(2\beta^2 + 2\beta + 1\right)}$$

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Challenge Problems

P42.89 (a) Let r represent the distance between the electron and the positron. The two move in a circle of radius $r/2$ around their center of mass with opposite velocities. The total angular momentum of the electron-positron system is quantized according to

$$L_n = \frac{mvr}{2} + \frac{mvr}{2} = n\hbar \qquad \text{where} \quad n = 1, 2, 3, \ldots$$

For each particle, $\sum F = ma$ expands to

$$\frac{k_e e^2}{r^2} = \frac{mv^2}{r/2}$$

We can eliminate $v = \frac{n\hbar}{mr}$ to find

$$\frac{k_e e^2}{r} = \frac{2mn^2\hbar^2}{m^2 r^2}$$

So the separation distances are

$$r = \frac{2n^2\hbar^2}{mk_e e^2} = 2a_0 n^2$$

Comparing this result to Equations 42.10 and 42.11, we see the allowed separation distances are two times the allowed radii of the Bohr hydrogen atom. Therefore,

$\boxed{r_n = 0.106n^2, \text{ where } r_n \text{ is in nanometers and } n = 1, 2, 3, \ldots}$

(b) The orbital radii are $\frac{r}{2} = a_0 n^2$, the same as for the electron in hydrogen. The energy can be calculated from

$$E = K + U = \frac{1}{2}mv^2 + \frac{1}{2}mv^2 - \frac{k_e e^2}{r}$$

Since $mv^2 = \frac{k_e e^2}{2r}$,

$$E = \frac{k_e e^2}{2r} - \frac{k_e e^2}{r} = -\frac{k_e e^2}{2r} = \frac{-k_e e^2}{4a_0 n^2}$$

Comparing this result to Equations 42.13 and 42.14, we see the allowed energies are half those of the Bohr hydrogen atom. Therefore,

$\boxed{E_n = -\frac{6.80}{n^2}, \text{ where } E_n \text{ is in electron volts and } n = 1, 2, 3, \ldots}$

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P42.90 (a) Suppose the atoms move in the $+x$ direction. The absorption of a photon by an atom is a completely inelastic collision, described by

$$mv_i\hat{\mathbf{i}} + \frac{h}{\lambda}\left(-\hat{\mathbf{i}}\right) = mv_f\hat{\mathbf{i}} \quad \text{so} \quad v_f - v_i = -\frac{h}{m\lambda}$$

This happens promptly every time an atom has fallen back into the ground state, so it happens every $10^{-8}\ \text{s} = \Delta t$. Then,

$$a = \frac{v_f - v_i}{\Delta t} = -\frac{h}{m\lambda\Delta t} \sim -\frac{6.626\times10^{-34}\ \text{J}\cdot\text{s}}{\left(10^{-25}\ \text{kg}\right)\left(500\times10^{-9}\ \text{m}\right)\left(10^{-8}\ \text{s}\right)}$$

$$\sim \boxed{-10^{6}\ \text{m/s}^2}$$

(b) With constant average acceleration,

$$v_f^2 = v_i^2 + 2a\Delta x \qquad 0 \sim \left(10^{3}\ \text{m/s}\right)^2 + 2\left(-10^{6}\ \text{m/s}^2\right)\Delta x$$

so $$\Delta x \sim \frac{\left(10^{3}\ \text{m/s}\right)^2}{10^{6}\ \text{m/s}^2}\ \boxed{\sim 1\ \text{m}}.$$

P42.91 (a) From Equation 42.13, the allowed energies are $E_n = \frac{k_e e^2}{2a_0}\left(\frac{1}{n^2}\right)$, where, from Equation 42.11, the Bohr radius is

$$a_0 = \frac{\hbar^2}{m_e k_e e^2} = \frac{\left(h/2\pi\right)^2}{m_e k_e e^2} = \frac{h^2}{4\pi^2 m_e k_e e^2}$$

Combining these gives

$$E_n = \frac{k_e e^2}{2}\frac{4\pi^2 m_e k_e e^2}{h^2}\left(\frac{1}{n^2}\right) = \frac{2\pi^2 m_e k_e^2 e^4}{h^2}\left(\frac{1}{n^2}\right)$$

For a transition from state n to state $n-1$,

$$hf = \Delta E = \left(\frac{2\pi^2 m_e k_e^2 e^4}{h^2}\right)\left(\frac{1}{(n-1)^2} - \frac{1}{n^2}\right)$$

$$hf = \Delta E = \left(\frac{2\pi^2 m_e k_e^2 e^4}{h^2}\right)\frac{n^2 - \left(n^2 - 2n + 1\right)}{n^2(n-1)^2}$$

which gives

$$f = \left(\frac{2\pi^2 m_e k_e^2 e^4}{h^3}\right)\frac{2n-1}{n^2(n-1)^2}$$

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(b) As $n \to \infty$, we find the quantum result:

$$f \to \frac{2\pi^2 m_e k_e^2 e^4}{h^3}\frac{2}{n^3} = \frac{4\pi^2 m_e k_e^2 e^4}{h^3 n^3}$$

The classical frequency is $f = \dfrac{v}{2\pi r}$, where classically, from Equation 42.8, $v^2 = \dfrac{k_e e^2}{m_e r}$. By substituting, the relation for the classical frequency becomes

$$f = \frac{v}{2\pi r} = \frac{1}{2\pi r}\sqrt{\frac{k_e e^2}{m_e r}} = \sqrt{\frac{k_e e^2}{4\pi^2 m_e r^3}}$$

From Equation 42.10, the radius $r = r_n = \dfrac{n^2\hbar^2}{m_e k_e e^2} = \dfrac{n^2 h^2}{4\pi^2 m_e k_e e^2}$; substituting this yields

$$f = \sqrt{\frac{k_e e^2}{4\pi^2 m_e r^3}} = \sqrt{\frac{k_e e^2}{4\pi^2 m_e}\left(\frac{4\pi^2 m_e k_e e^2}{n^2 h^2}\right)^3}$$

$$= \sqrt{\frac{\left(4\pi^2\right)^2 m_e^2 k_e^4 e^8}{n^6 h^6}} = \frac{4\pi^2 m_e k_e^2 e^4}{h^3 n^3}$$

The classical frequency is $4\pi^2 m_e k_e^2 e^4 / h^3 n^3$. We see that the Bohr result for large n reduces to the classical result.

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P42.2 (a) 1 875 nm, 1 282 nm, 1 094 nm; (b) infrared

P42.4 (a) $\lambda_{mn} = \left| \dfrac{1}{1/\lambda_{m1} - 1/\lambda_{n1}} \right|$; (b) $k_{mn} = |k_{m1} - k_{n1}|$

P42.6 (a) See P42.6(a) for full explanation; (b) 0.846 ns

P42.8 See P42.8 for full explanation.

P42.10 (a) 2.86 eV; (b) 0.472 eV

P42.12 (a) 1.89 eV; (b) 656 nm; (c) 3.40 eV; (d) 365 nm; (e) 365 nm

P42.14 4.42×10^4 m/s

P42.16 (a) 97.3 nm; (b) 1.88 μm; (c) infrared; (d) Paschen; (e) 97.3 nm; (f) ultraviolet; (g) Lyman

P42.18 (a) 0.212 nm; (b) 9.97×10^{-25} kg · m/s; (c) 2.11×10^{-34} kg · m^2/s; (d) 3.40 eV; (e) –6.80 eV; (f) –3.40 eV

P42.20 (a) 152 as; (b) 8.23×10^9 revolutions; (c) Its lifetime in electron years is comparable to the lifetime of the Sun in Earth years, so we can think of it as a long time.

P42.22 (a) 1.31 μm; (b) 164 nm

P42.24 (a) $\dfrac{\hbar}{2r}$; (b) $\dfrac{\hbar^2}{2m_e r^2}$; (c) $\dfrac{\hbar^2}{2m_e r^2} - \dfrac{k_e e^2}{r}$; (d) $\dfrac{\hbar^2}{m_e k_e e^2} = a_0$; (e) –13.6 eV; (f) We find our results are in agreement with the Bohr theory.

P42.26 (a) See P42.26(a) for full explanation; (b) $E = -\dfrac{k_e e^2}{2a_0}$

P42.28 (a) 1; (b) 0.497

P42.30 (a) See P42.30(a) for a list of all sets; (b) See P42.30(b) for a list of all sets.

P42.32 (a) $\sqrt{6}\hbar$; (b) L_z can have the values $-2\hbar$, $-\hbar$, 0, $\hbar$ and $2\hbar$; (c) 145°, 114°, 90.0°, 65.9°, and 35.3°

P42.34 (a) 2; (b) 8; (c) 18; (d) 32; (e) 50

P42.36 (a) 3.99×10^{17} kg/m^3; (b) 8.17 am; (c) 1.77 Tm/s; (d) It is $5.91 \times 10^3 c$, which is huge compared with the speed of light and impossible.

P42.38 The electron energy is not enough to excite the hydrogen atom from its ground state to even the first excited state.

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P42.40 (a) the 4*s* subshell; (b) We would expect [Ar]$3d^4 4s^2$ to have lower energy, but [Ar]$3d^5 4s^1$ has more unpaired spins and lower energy according to Hund's rule; (c) chromium

P42.42 See P42.42 for the complete table.

P42.44 1*s*, 2*s*, 2*p*, 3*s*, 3*p*, 4*s*, 3*d*, 4*p*, 5*s*, 4*d*, 5*p*, 6*s*, 4*f*, 5*d*, 6*p*, 7*s*

P42.46 $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 4f^{14} 5s^2 5p^6 5d^{10} 5f^{14} 6s^2 6p^6 6d^8 7s^2$

P42.48 See P42.48 for full explanation.

P42.50 0.310 nm

P42.52 (a) For the 3*p* state, 1.4 and for the 3*d* state, 1.0; (b) See P42.52(b) for full explanation.

P42.54 L shell = 11.8 keV, M shell = 10.2 keV, N shell = 2.47 keV

P42.56 (a) $\dfrac{1240\ \text{V}\cdot\text{nm}}{\lambda}$; (b) The potential difference is inversely proportional to the wavelength; (c) Yes. It predicts a minimum wavelength of 33.5 pm when the accelerating voltage is 37 keV, in agreement with the minimum wavelength in the figure; (d) Yes, but it might be unlikely for a very high energy electron to stop in a single interaction to produce a high-energy gamma ray, and it might be difficult to observe the very low intensity radio waves produced as bremsstrahlung by low-energy electrons; (e) The potential difference goes to infinity as the wavelength goes to zero; (f) The potential difference goes to zero as the wavelength goes to infinity.

P42.58 633 nm

P42.60 (a) 18.37 eV; (b) 217 nm; (c) 93.1 nm

P42.62 (a) 1.26×10^{-33}; (b) -1.15×10^6 K; (c) A population inversion cannot happen in thermal equilibrium.

P42.64 (a) $\lambda_1 = 632.809\ 14$ nm, $\lambda_2 = 632.808\ 57$ nm, $\lambda_3 = 632.809\ 71$ nm, three; (b) 697 m/s (c) See P42.64(c) for full description.

P42.66 0.389 T/m

P42.68 (a) 1.63×10^{-18} J; (b) 7.88×10^4 K

P42.70 5.39 keV

P42.72 See P42.72 for full explanation.

P42.74 In order to have a maximum value of m_ℓ equal to 3, we need to have $\ell = 3$, which is inconsistent with the first result.

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P42.76 (a) See P42.76(a) for full explanation; (b) 2.53×10^{74}; (c) 1.18×10^{-63} m; (d) This number is *much smaller* than the radius of an atomic nucleus ($\sim 10^{-15}$ m), so the distance between quantized orbits of the Earth is too small to observe.

P42.78 (a) $\left(\frac{2r^2}{a_0^2} + \frac{2r}{a_0} + 1\right) e^{-2r/a_0}$; (b) See ANS. FIG. P42.78; (c) 1.34 a_0

P42.80 (a) –8.16 eV, –2.04 eV, –0.902 eV, –0.508 eV, –0.325 eV; (b) 1 090 nm, 811 nm, 724 nm, and 609 nm; (c) 122 nm, 103 nm, 97.3 nm, 95.0 nm, 91.2 nm; (d) The spectrum could be that of hydrogen, Doppler-shifted by motion away from us at speed 0.471c.

P42.82 (a) Al: 2.55×10^{-10} m ~ 10^{-1} nm and U: 2.76×10^{-10} m ~ 10^{-1} nm; (b) The outermost electron in any atom sees the nuclear charge screened by all the electrons below it. If we can visualize a single outermost electron, it moves in the electric field of net charge $+Ze - (Z - 1)e = +e$, the charge of a single proton, as felt by the electron in hydrogen. So the Bohr radius sets the scale for the outside diameter of every atom. An innermost electron, on the other hand, sees the nuclear charge unscreened, and the scale size of its (K-shell) orbit is a_0/Z.

P42.84 (a) $c\Delta t$; (b) $\frac{\lambda T_{\text{ER}}}{hc}$; (c) $\frac{4\lambda T_{\text{ER}}}{\pi hc^2 d^2 \Delta t}$

P42.86 See ANS. FIG P42.86 for the energy-level diagram.

P42.88 $e^{-2\beta}\left(2\beta^2 + 2\beta + 1\right)$

P42.90 (a) -10^6 m/s^2; (b) ~1 m

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